Section 7 of "25 Years of SMOTE": Empirical Comparisons and Benchmarks

Draft v4.1, 8 November 2026

SMOTE25 team (Section 7: Tan)

7 Empirical Comparisons and Benchmarks

Every method in the SMOTE family is judged in the same way. Its authors resample a set of benchmark datasets, train one or more classifiers on the result, score the classifiers on held-out data with a few measures, compare the scores with those of competing methods, and, in most recent papers, test whether the differences are larger than chance. The method itself is never measured; only its effect on a downstream classifier is. Each choice in this chain, from the datasets to the statistical test, can change which method comes out ahead, and the choices have changed a great deal since SMOTE was introduced in 2002 (Chawla et al., 2002).

This section is written for a reader who is about to propose or compare a SMOTE-family method and wants to know how such a study is evaluated today, why it is evaluated that way, and what a reviewer can reasonably expect. We read the evaluation of all 71 empirical studies in our corpus and follow the chain in order. After the corpus and the way we read it (Section 7.1) and a comparison of SMOTE's own evaluation with today's (Section 7.2), we take each link in turn: the benchmark data and the task (Section 7.3), the base learners (Section 7.4), the competitors and baselines (Section 7.5), the performance measures, including why G-mean became the most reported one and how it differs from F1 (Section 7.6), evidence that does not come from a classifier (Section 7.7), the validation protocol and the choice of settings (Section 7.8), statistical testing (Section 7.9) and reporting (Section 7.10). We then ask why evaluation practice changed when it did (Section 7.11) and what the comparisons conclude (Section 7.12), collect the recommendations into a protocol for a new study (Section 7.13), and state the limits of our analysis (Section 7.14). Appendix A describes the illustrative runs we use in two places and gives the coding of every study.

The short answers are these. SMOTE was evaluated with three learners and three quantities (AUC, the true positive rate and the true negative rate). A typical study published since 2021 still uses three learner families and three measures, but studies no longer agree on which: the 24 recent studies draw on 12 learner families and 21 measures and use 19 different combinations of measures. Most of the change in learners follows a change of toolkit, from Weka and KEEL to Python and scikit-learn, and most of the change in measures follows the absorption of single-class rates into composite measures, the growth of benchmarks and the growth of multi-class work. G-mean is now the most reported measure because it measures exactly what resampling changes, the balance between the true positive and true negative rates, and for the same reason it can make any shift of the decision threshold look like an improvement. Every study evaluates through classification, direct checks of the resampled data are rare (6 studies), simple non-resampling baselines such as class weighting are almost never included, and the weakest step of the typical evaluation is the choice of settings, which in half of the studies is made on the data used to report results.

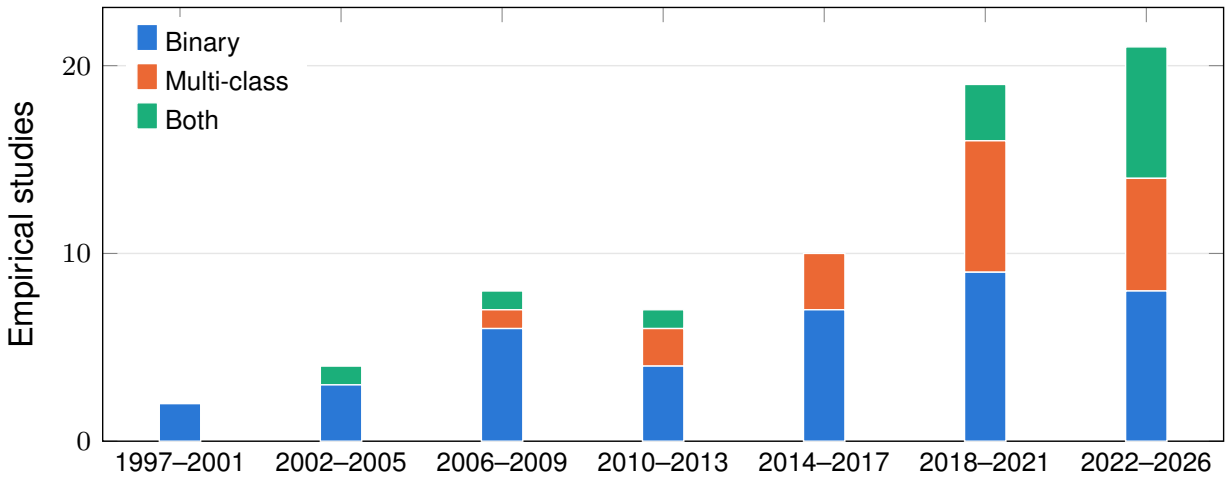


Figure 1: Empirical Studies in the Corpus by Publication Period and Problem Setting

7.1 Corpus, Coding and How to Read the Numbers

The corpus is the SMOTE25 collection selected in Section 3: 76 papers after duplicate copies were merged. Of these, 71 report an empirical evaluation on imbalanced data; the other five are two surveys (Pradipta et al., 2021; Hassanat et al., 2022) and three foundational works that predate imbalance-aware evaluation (Wilson, 1972; Tomek, 1976; Freund and Schapire, 1997). Among the 71 empirical studies, 39 evaluate binary problems, 19 multi-class problems and 13 both (Figure 1). Thirty-seven of the 71 are SMOTE-family methods in the sense of Section 4 (34 core and 3 partial members: 24 binary only, 5 multi-class only and 8 for both settings); the others are competing resamplers, ensembles, cost-sensitive methods, generative models and empirical studies that SMOTE-family papers compare with. We report results for all 71 and point out where the 37 family members behave differently.

To show change over time we group the studies into four periods of similar size: 1997 to 2009 (14 studies), 2010 to 2015 (13), 2016 to 2020 (20) and 2021 to 2026 (24). A statement such as "G-mean rose from 29% to 71%" means that 4 of the 14 studies of the first period and 17 of the 24 studies of the last period report G-mean. With periods this small, one study moves a share by 4 to 8 points, so we only call a change a trend when it exceeds 15 points or when a practice appears or disappears completely.

For each study we coded, from the full text and supplementary files, the benchmark data, the base learners, the competing methods, every measure together with its exact definition, the validation protocol, how hyperparameters were chosen, the statistical tests and the form in which results are reported. A field the paper does not state was coded as not reported, never inferred. A second reading recorded why each study made its choices: the reasons it gives for its data, learners and measures, the measure and the data used to choose its settings, the methodological sources it cites, and any evidence about the resampled data that does not come from a classifier. Every code from the second reading carries a page reference and a short verbatim quote, so it can be checked against the paper [TODO: a co-author to verify the second-reading quotes before submission]. We also recorded which software a study names (for example Weka, KEEL, MATLAB, scikit-learn or imbalanced-learn). Only named software is counted, so these counts are lower bounds. Table 31 in the appendix lists the main codes of every study.

When we explain why a practice changed, we use three levels of evidence and say which one

applies. Direct evidence means that the studies themselves give the reason, or that the link can be measured in the corpus, for example the share of composite-measure users who still report recall separately. Circumstantial evidence means that timing and association point the same way but no study states the link, for example the parallel rise of Python and of CART. A conjecture is a mechanism we propose that the corpus cannot confirm. None of these is a causal estimate in the statistical sense; with 71 studies and no intervention, the corpus can show that a change and its proposed cause move together and that the studies say so, not more.

To see whether careful evaluation is rewarded, we took citation counts from OpenAlex on 18 October 2026, available for 70 of the 71 studies, and compare the median number of citations of the studies that follow a practice with that of the studies that do not (Section 7.11). We report the raw counts. A count has to be read against the year of publication, because older papers have had more time to be cited: the median study published up to 2009 has 1,738 citations, against 21 for the studies published since 2021. Every comparison therefore gives the median year of each group next to its median count, and where the two groups differ in age we also compare studies of the same period.

Relation to the fifteen-year review. This analysis complements the review that marked the fifteenth anniversary of SMOTE (Fernández et al., 2018). That review catalogued more than 85 extensions by seven properties of their design (how the instances to oversample are selected, integration with undersampling, the type of interpolation, operation under changes of dimensionality, adaptive generation, relabelling and the filtering of noisy synthetic instances), listed SMOTE-based ensembles and 22 SMOTE-based methods for 12 other learning paradigms, and set out the open challenges: small disjuncts, noise, class overlap, dataset shift, dimensionality, real-time processing and big data. On evaluation it says three things. It recalls why accuracy failed on the data that motivated SMOTE, where a decision tree reached about 97% accuracy on the mammography data while always predicting the majority class gave 97.68%, and the two kinds of error had different costs, so the trade-off between true and false positives had to be shown with ROC analysis. It summarises the empirical comparisons available in 2018. And it recommends a validation scheme that is robust to the dataset shift induced by partitioning. It does not examine which measures, protocols or tests the studies use, which is what this section adds. Table 1 lists the points of the review that bear on evaluation and what the studies published since did; we return to each of them below (Sections 7.5, 7.7, 7.8 and 7.11).

7.2 From SMOTE (2002) to Today: Learners and Measures

What SMOTE measured. The original paper evaluated SMOTE on nine datasets with three learners, the C4.5 decision tree, the Ripper rule learner and naive Bayes, chosen so that SMOTE could be compared with methods that handle misclassification costs directly (Chawla et al., 2002). Each dataset was split by a single 10-fold cross-validation. For every amount of oversampling the paper recorded two rates, the percentage of minority examples classified correctly (the true positive rate, TPR) and the percentage of majority examples classified incorrectly (the false positive rate, one minus the true negative rate, TNR), drew the resulting points as ROC curves, and summarised each curve by the area under it (AUC). It also used the ROC convex hull to find potentially optimal classifiers and, for one dataset, reported the size of the pruned tree. The paper argued explicitly that accuracy cannot be used under imbalance, and it did not use G-mean, F1 or a significance test. In short, SMOTE was evaluated with three learners and three quantities (AUC, TPR and TNR) plus ROC plots, because what mattered in the problem that led to it was the trade-off between true and false positives (Fernández et al., 2018).

Table 1: Points of the Fifteen-Year Review That Bear on Evaluation, and What Followed

Point made by the review (Fernández et al., 2018)	What the studies of the corpus did
Preprocessing and cost-sensitive learning are equally effective; compare with both (López et al., 2012)	A cost-sensitive or instance-weighting competitor in 5 studies, 3 of them after 2018 (Fernandez et al., 2013; Yang et al., 2018; Yao and Lin, 2021; Xie et al., 2022; Yadav and Vijay, 2025); plain class weighting in one (Yadav and Vijay, 2025)
Partition-induced dataset shift distorts cross-validation on imbalanced data; use DOB-SCV (Moreno-Torres et al., 2012; López et al., 2014)	No study uses DOB-SCV
The main open question is how synthetic examples relate to small disjuncts, noise and overlap	A direct check of the resampled data in 5 of the 36 studies since 2019, against 1 of 35 before; 2 count safe, borderline or rare examples (Liu et al., 2020; Han et al., 2023)
SMOTE-based methods exist for 22 other paradigms, each with its own measures	One of them (INOS, time series) is in the corpus, evaluated as classification (Cao et al., 2013)
High-dimensional data and big data are open challenges (Blagus and Lusa, 2013)	No study evaluates on big data; image data appear in 4 studies, all from 2018 to 2020

Table 2: What SMOTE Used in 2002 and What Studies Use Now

	SMOTE (Chawla et al., 2002)	Studies published 2021 to 2026 ($n = 24$)
Learners	C4.5, Ripper (rule learner), naive Bayes	12 families (32 learner names as written); median 3 families per study (range 1 to 6), with 9 of 24 studies using a single family
Scalar measures	AUC, TPR (%TP, Acc^+), TNR (Acc^-, or 1 − %FP)	21 distinct measures; median 3 per study; 19 different combinations among 24 studies
Most used measures	AUC (from ROC curves)	G-mean (17 of 24), AUC (13), F1 or F-measure (10), MAUC (6)
Other evidence	ROC curves and ROC convex hull, tree size, a 2-D plot of decision regions	2-D plots of synthetic samples (15 of 24), runtime (7), direct checks of the resampled data (2)
Data and protocol	9 datasets, single 10-fold CV, no significance test	KEEL data in 67%, repeated CV in 62%, a significance test in 83%
Code	not released	released by 12 of 24

What studies use now. Table 2 sets this against the 24 studies published from 2021 to 2026. Two numbers have hardly moved. The median study uses 3 learner families (range 1 to 6), and 9 of the 24 studies use a single family; it also reports a median of three scalar measures. What has changed is how much the studies differ from each other. Together they use 12 learner families and 21 different measures, and among the 24 studies there are 19 different combinations of measures; only three combinations occur more than once. The 12 families are written in the papers under 32 different names, because the same algorithm is named with its kernel, library or settings ("SVM", "SVM (RBF)", "Linear SVM (LIBSVM)"); Table 3 lists the algorithms inside each family.

The most used family since 2021 is the support vector machine (13 of the 24 studies, 54%), followed by the CART decision tree (12, 50%) and k-nearest neighbours (9, 38%). The SVM leads for two reasons that the corpus shows directly. It is the one strong learner that both generations of toolkits offer, as SMO and LIBSVM in Weka and as SVC in scikit-learn, so it survived the change of software described below. And it has become the default choice of studies that test with a single learner: 4 of the 9 single-learner studies since 2021 use an SVM, against 3 that use C4.5, the single learner of earlier work; one of them calls the SVM the most well-known and straightforward classification algorithm (Palli et al., 2022). CART is second because it is the decision tree of scikit-

Table 3: Learner Families Used by the 24 Studies Published 2021 to 2026

Family	Algorithms named in the studies	Studies	Used alone
Support vector machine	SVM with RBF or linear kernel (LIBSVM, scikit-learn), one-vs-rest for several classes	13 (54%)	4
CART (decision tree)	CART, the decision tree of scikit-learn; also the base tree of some ensembles	12 (50%)	1
k-NN	k-nearest neighbours, usually $k = 5$	9 (38%)	–
Multilayer perceptron	multilayer perceptron trained by back-propagation	6 (25%)	–
Random forest	random forest	6 (25%)	–
C4.5 or C5.0	C4.5 (Weka J48, sometimes unpruned with Laplace smoothing) and C5.0	5 (21%)	3
Logistic regression	logistic regression	5 (21%)	–
Naive Bayes	naive Bayes, including Gaussian naive Bayes	5 (21%)	–
Gradient boosting	XGBoost, CatBoost, GBDT, HistGradientBoosting	5 (21%)	1
AdaBoost-type boosting	AdaBoost	2 (8%)	–
LDA	linear discriminant analysis	1 (4%)	–
Other	dynamic selection of one-vs-one classifiers (DOVO) over several learners	1 (4%)	–

Used alone: studies whose only learner is of this family (9 of the 24). The papers write these algorithms under 32 different names, for example “SVM, “SVM (RBF) and “Linear SVM (LIBSVM).

learn, and studies that move to Python take its tree with it.

A reader who wants to compare the numbers of two recent papers will usually find that they do not share a measure, a learner or both. The variety itself is not a problem, because different applications need different measures, but it means that the evidence for a method cannot be pooled across papers. This is the first thing a new study should plan for: if its results are to be compared with earlier work, it has to report the measures and learners that earlier work used, in addition to the ones it prefers.

What survived from 2002. Of the three learners SMOTE used, C4.5 (5 of the 24 studies since 2021) and naive Bayes (5) are still in use, and Ripper is not. Of its three quantities, AUC is still reported by more than half of the recent studies (13 of 24) and TPR by 5, while TNR is no longer reported on its own and no recent study draws ROC curves. TNR has not disappeared from the evaluations, though; it is inside G-mean. The additions are G-mean, F1 and its multi-class variants, CART, support vector machines, logistic regression, random forests and gradient boosting, and a family of multi-class measures (MAUC, MG, AvAcc, CBA, CEN, macro-F1). The 37 SMOTE-family studies follow the same pattern, with a split by setting: the 24 binary-only methods report AUC most often (18 of 24), then G-mean (14), recall (12) and F1 (12), while the methods for multi-class problems report G-mean or MG first (4 of the 5 multi-class-only methods and 5 of the 8 that cover both settings), with MAUC and AvAcc next.

Why the learners changed. Table 4 lists every learner family with its share in each period and the cause of its rise or fall. One cause explains most of the table: the field changed its software. Before 2016, studies that name their software name Weka (one study before 2010 and six from 2010 to 2015) or MATLAB; no study names Python. From 2016 the order reverses: 6 studies name Python or a Python library such as scikit-learn or imbalanced-learn in 2016 to 2020 and 12 since 2021, while Weka is named by 4 studies in 2016 to 2020 and by only one since 2021 (Figure 2). The learners followed the software. C4.5 is the default tree of Weka (J48) and of KEEL, while the decision tree of scikit-learn is CART, and scikit-learn has no C4.5 and no rule learner. Rule learners,

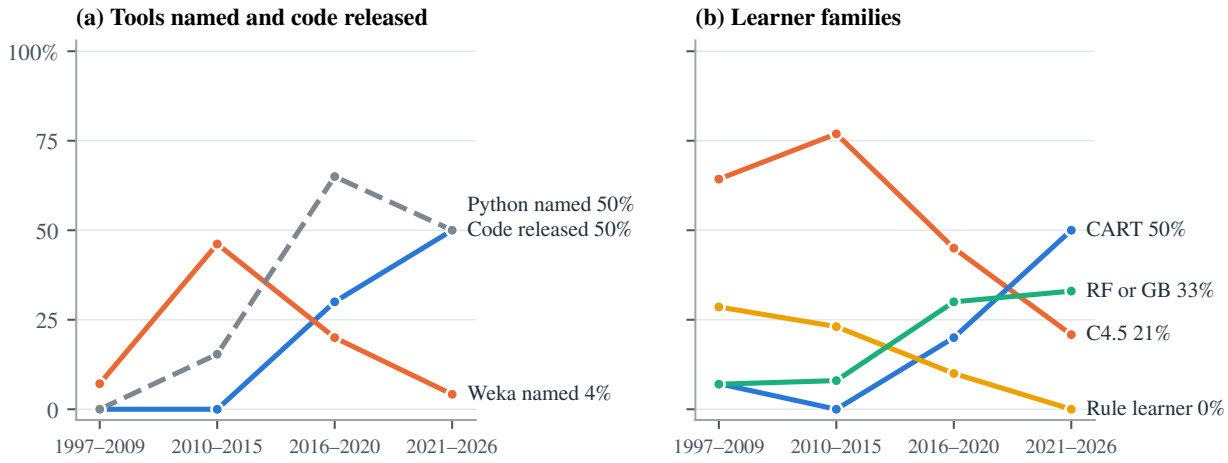


Figure 2: The Change of Toolkit and the Learners That Followed It

such as Ripper and PART, output an ordered list of if–then rules: Ripper grows and prunes one rule at a time, and PART reads rules off partial C4.5 trees. Accordingly C4.5 fell from 64% of the studies before 2010 to 21% since 2021, CART rose from 7% to 50%, and rule learners fell from 29% to none; the last study to use one is SOUP (2019) (Janicka et al., 2019). The users of the two trees show the same split: 11 of the 33 C4.5 studies name Weka and 2 name Python, while 10 of the 17 CART studies name Python and only one names Weka (Table 10 in Section 7.4). Code release, which also became common with Python, separates them as well: 13 of the 17 CART studies release code, against 8 of the 33 C4.5 studies. The tree did not disappear; it changed with the toolkit, and so did the algorithm inside it. A consequence for a reader is that C4.5 results in older papers and CART results in newer ones are not interchangeable, because the two trees split and prune differently.

The other learners fit the same pattern. Learners that exist in both toolkits survived the change: k-nearest neighbours (k-NN), support vector machines and naive Bayes. Logistic regression arrived with Python (6 of its 13 users name Python and none names Weka) and is chosen, in the words of one study, because a learner with few hyperparameters is less likely to bias the comparison through its configuration (Douzas et al., 2018). Random forests and gradient boosting rose from 7% to 33%. Here the reason the studies give is realism: these are the learners practitioners use, and a resampler that only helps weak learners is of limited value (Vuttipittayamongkol et al., 2018; Vuttipittayamongkol and Elyan, 2020; Yadav and Vijay, 2025). AdaBoost-type boosting, which represented ensembles from 2016 to 2020, has since been replaced by gradient boosting. Two families moved for reasons unrelated to software. The multilayer perceptron is carried by one line of work: all four studies of the MC-CCR line use it (Koziarski et al., 2020; Krawczyk et al., 2020; Liu et al., 2023; Ma et al., 2024), and one of them states that it adopts the learners, folds and measures of the first two. Convolutional networks appear only in the two image studies of 2018 and 2019 (Buda et al., 2018; Mullick et al., 2019) and vanish with image data, not because they were abandoned.

Why the measures changed. Table 5 does the same for every measure. Four causes account for nearly all of it. The first is absorption: single-class rates were not abandoned but folded into composite measures that contain them. G-mean contains TPR and TNR, and F1 contains recall and precision. As the composites spread, studies stopped reporting their parts separately. Among

Table 4: Base Learners by Period and Why Their Use Changed

Learner	–2009 (14)	–2015 (13)	–2020 (20)	–2026 (24)	Trend	Driver	Cause of the change
C4.5 or C5.0	64%	77%	45%	21%	falling	T, L	Shipped with Weka (J48) and KEEL, the tools of the first decade; scikit-learn has no C4.5. Users name Weka in 11 of 33 studies and Python in 2. The five users since 2021 keep it to stay comparable with earlier work in their line.
CART (decision tree)	7%	0%	20%	50%	rising	T	The decision tree of scikit-learn is CART, so the tree did not disappear: it changed with the toolkit. 13 of 17 users release code (C4.5: 8 of 33), and 12 of the 17 are from 2021 onwards.
SVM	21%	46%	30%	54%	rising	T, L	Available in both toolkits, and the most frequent choice of studies that use one learner only: 4 of the 9 single-learner studies since 2021 use an SVM, 3 use C4.5.
k-NN	14%	46%	55%	38%	rising	T, R	Available in both toolkits and chosen to cover different learning paradigms (11 of 28 say so). Its share fell after 2020 because studies still use about three learners while SVM, RF and boosting took more of the slots.
Multilayer perceptron	7%	31%	30%	25%	rising	L	Carried by one line of work: all four studies of the MC-CCR line (MC-CCR, MC-RBO, NROMM, MC-MBRC) use it; three early studies used it alone.
Naive Bayes	14%	23%	20%	21%	stable	L	Cheap, parameter-free and present in every toolkit; never used alone (0 of 14), so nothing pushed it up or down.
Logistic regression	7%	8%	30%	21%	stable	T, R	Arrived with Python (6 of 13 users name Python, none name Weka); chosen because few hyperparameters limit tuning bias.
Random forest	7%	8%	20%	25%	rising	R, T	Studies want to know whether resampling still helps a strong learner that practitioners use; RF is reported as top performing or most widely used.
Gradient boosting	0%	0%	15%	21%	new	R, T, L	The strongest off-the-shelf tabular learner today; 7 of 8 users name Python and their median benchmark has 34.5 datasets. QC-SMOTE keeps CatBoost to match a published benchmark.
AdaBoost-type boosting	7%	15%	25%	8%	stable	T, R	Represented the ensemble paradigm in 2016 to 2020 (5 studies) and was then replaced by gradient boosting (2 against 5 studies since 2021).
Rule learners (RIPPER, PART, MODLEM)	29%	23%	10%	0%	gone	T, L	Only in Weka or in the authors' code; scikit-learn has none. They were used by two groups (Chawla and co-authors early on; the Poznań group in SPIDER2 and SOUP) and not after SOUP (2019).
Convolutional network	0%	0%	10%	0%	gone	P	Not abandoned: the two image studies (2018, 2019) used CNNs because the data were images, and no study since 2021 uses image data.
LDA	0%	0%	5%	4%	new	N	Used twice because it has no parameters to tune.
Other	0%	0%	5%	4%	new	N	Method-specific choices: a Bayesian KDE classifier that matches the generator of CGMOS, and a one-vs-one ensemble in RSMOTE.

Columns give the share of the studies of each period (number of studies in brackets). Trend: gone = no study since 2021; new = first used in 2016 or later; rising or falling = the share changed by 15 points or more between the first and the last period. Drivers: T = change of toolkit; L = following a line of earlier work; R = realism or a critique; P = change of problem or data type; N = used in at most two studies. Tool names count only studies that name the tool.

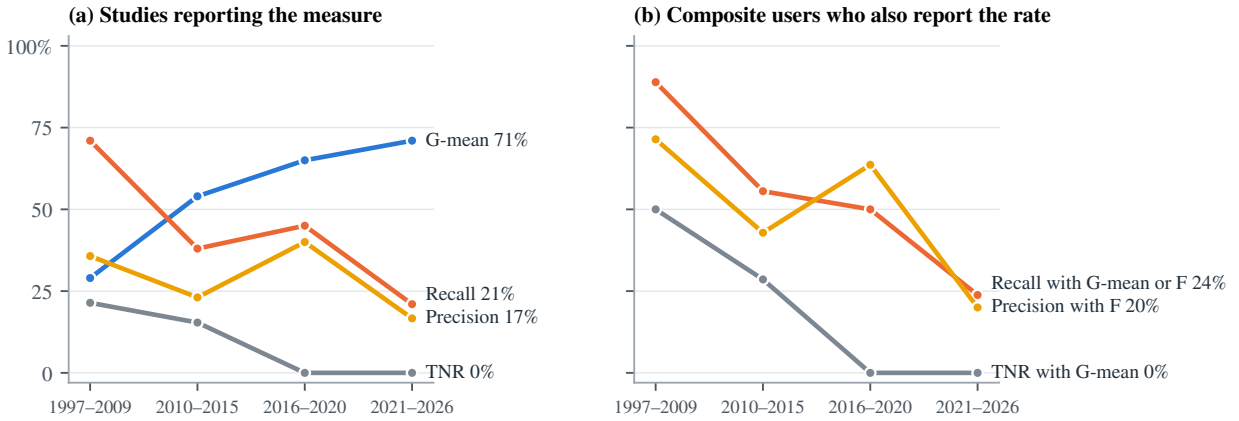


Figure 3: Single-Class Rates Absorbed into Composite Measures

the studies that report G-mean or F1, the share that also reports recall on its own fell from 8 of 9 before 2010 to 5 of 21 since 2021; among the F1 users, the share that also reports precision fell from 5 of 7 to 2 of 10; and among the G-mean users, the share that also reports TNR fell from 2 of 4 to none of 17 (Figure 3). This is why recall fell from 71% to 21% while G-mean rose from 29% to 71%: the same quantity moved inside another one.

The second cause is scale. ROC curves and per-dataset confusion matrices can be read for a handful of datasets, not for dozens. The median benchmark grew from 6.5 datasets per study before 2010 to 23 since 2021, and the studies that drew ROC curves used a median of 12 datasets, against 34 for studies that report AUC without curves. The curves gave way to the area under them, and the last study to draw them is an image study with three datasets (Buda et al., 2018). Larger benchmarks and heavier methods are also why running time is reported more often (29% since 2021, against 14% before).

The third cause is a change of problem. Multi-class studies grew from 3 in the first period to 15 in the last, and binary measures do not apply to them directly. MAUC, the multi-class extension of AUC by Hand and Till (Hand and Till, 2001), is reported only by multi-class studies, and nine studies cite Hand and Till when they justify their measures. Macro-F1, the class-averaged F1 that scikit-learn offers as a standard option, follows the same path. Generative studies, which entered the corpus in 2019, brought their own tradition, in which a classifier is trained on synthetic data and tested on real data and scored by accuracy and F1 (Xu et al., 2019; Kim et al., 2022; Ren et al., 2025, 2026); four of the five studies that still report accuracy since 2016 are generative studies.

The fourth cause is the line of work a study belongs to. CBA and CEN entered this literature with MC-CCR and MC-RBO in 2020 (Koziarski et al., 2020; Krawczyk et al., 2020), and every later study that reports them is from the same line and uses the same KEEL multi-class benchmarks. AvAcc came from the KEEL group in 2013 (Fernandez et al., 2013) and travelled with them. Measures of this kind spread because a study that wants to be compared with its predecessors reports what they reported.

Two measures moved for other reasons. MCC appeared in 2020 after critiques of F1, which ignores true negatives, and of accuracy (Boughorbel et al., 2017); all four of its users name Python and release code, where MCC is a standard function. PR-AUC, by contrast, did not grow even though its best-known recommendation (Saito and Rehmsmeier, 2015) has more than 5,000 citations. Its share stayed at 8 to 10%. We can only conjecture why: PR-AUC needs well-ranked scores, which C4.5 and rule learners do not produce; it depends on the class prior, so averages over benchmarks

with very different imbalance ratios are hard to read; and no benchmark or toolkit in this literature reports it by default. The last explanation is consistent with Section 7.11, where every practice that did spread had a tool or a benchmark behind it. Table 25 in that section summarises the evidence for each of these drivers by period.

Table 5: Measures by Period and Why Their Use Changed

Measure	–2009	–2015	–2020	–2026	Trend	Driver	Cause of the change
Overall							
Accuracy, error rate	21%	38%	10%	12%	stable	P	Criticised since 1997 and used as a secondary number; since 2016 four of its five users are generative studies (CTGAN, SOS, GOIO, DGOT) that follow the train-on-synthetic, test-on-real tradition.
Single-class rates							
Recall (TPR)	71%	38%	45%	21%	falling	A	Absorbed into G-mean and F1: among studies reporting one of them, the share that also reports recall separately fell from 8 of 9 to 5 of 21.
Precision	36%	23%	40%	17%	falling	A, P	Absorbed into F1 (5 of 7 F users also report it before 2010, 2 of 10 since 2021); what remains is per-class precision in multi-class studies.
Specificity (TNR)	21%	15%	0%	0%	gone	A	Absorbed into G-mean: 2 of 4 G-mean users report it before 2010, 0 of 17 since 2021; last reported alone in 2013.
Threshold composites							
G-mean	29%	54%	65%	71%	rising	A, L	Measures the TPR–TNR trade-off that resampling moves, ignores the class prior, needs only labels and extends to multi-class (MG); see Section 7.6.1.
F1	43%	31%	35%	29%	stable	R	Inherited from information retrieval and kept by studies that focus on the positive class; it does not grow because it depends on the class prior, which makes averages over KEEL benchmarks hard to read (Section 7.6.2).
F-measure (β not stated)	7%	23%	20%	12%	stable	R	The same measure as F1 named F-measure or F-value without β; read it together with F1 (35 studies in total).
MCC	0%	0%	5%	12%	new	R, T	Follows critiques of F1 and accuracy; all four users name Python and release code.
Balanced accuracy	0%	0%	5%	8%	new	T	The arithmetic mean of TPR and TNR, available in standard toolkits; three studies.
Threshold-free ranking							
ROC-AUC	36%	62%	35%	54%	rising	L, P	Standard for binary studies on KEEL data; its share drops when multi-class studies, which use MAUC instead, are frequent (2016 to 2020) and recovers afterwards.
MAUC	7%	8%	20%	25%	rising	P	Every user is a multi-class study, and multi-class studies grew from 3 to 15 per period; one formula for the whole field.

Measure	–2009	–2015	–2020	–2026	Trend	Driver	Cause of the change
PR-AUC	0%	8%	10%	8%	stable	R	Recommended for imbalance but needs good scores, which C4.5 and rule learners do not give, and depends on the prior; 4 of 5 users name Python.
Multi-class aggregates							
Macro-F1	0%	0%	10%	21%	new	P, T	F1 averaged over classes, a standard option in scikit-learn; 6 of 7 users are multi-class studies.
AvAcc	0%	8%	15%	12%	stable	L	From the KEEL group (2013); adopted by the MC-CCR line as part of a fixed set (AvAcc, CBA, MG, CEN): 5 of 7 users belong to that line, all use KEEL data.
CBA	0%	0%	10%	17%	new	L	Introduced to this literature with MC-CCR and MC-RBO (2020); all 6 users belong to that line, one of them states it follows MC-CCR.
CEN	0%	0%	10%	12%	new	L	Comes with CBA in the same set; all 5 users belong to the MC-CCR line.
Macro-precision	0%	0%	5%	12%	new	L, P	Per-class precision averaged (Pavg) in the AMDO and EMDO line and in studies that compare with it.
Per-class recall	7%	0%	10%	4%	stable	P	Multi-class studies that inspect the smallest classes one at a time (4 studies).
Other							
Misclassification cost	7%	8%	0%	0%	gone	P	Needs known error costs, which benchmark data do not have; only the two cost-sensitive studies (2006, 2010).
Running time	14%	15%	15%	29%	rising	R	Newer methods are heavier (optimisation, generative models, certificates) and benchmarks larger, so authors must show the method is practical; QC-SMOTE reports about 39 times the cost of SMOTE against 209 for VS-SMOTE.
ROC curves (plots)	29%	15%	10%	0%	gone	S	Readable only for a few datasets: users have a median of 12 datasets against 34 for AUC users who draw none; the median benchmark grew from 6.5 to 23 datasets.
Confusion matrices	7%	8%	10%	0%	gone	S	Shown per dataset only when there are few datasets (median 10); summarised by measures when there are dozens.
Fidelity of synthetic data	0%	0%	10%	4%	new	P	Comes with methods that generate data from a model (CTGAN, GAMO) and with CISO; see Section 7.7.

Used in one or two studies, so no trend can be read: kappa (Random Balance 2015); G-score of precision and recall (CGMOS 2016); FPR (Cluster-SMOTE 2006, for ROC plots); minimum sensitivity (DSRBF 2011, AMDO 2018); AUCm (AMDO 2018, NROMM 2023); Pmin (NROMM 2023); weighted F1 (SOS 2022); macro-recall (RSMOTE 2021).

Same conventions as Table 4; A = absorbed into a composite measure; S = larger benchmarks. F1 and F-measure are coded apart only because papers name them differently.

What this means for a new study. The learners and measures of a study are mostly inherited from its software and from the papers it wants to be compared with. That is not wrong, but it

should be a decision rather than a default. A new study should (i) state its software and versions, because the same name can hide different algorithms; (ii) include at least one learner that is sensitive to imbalance, at least one strong ensemble such as a random forest or gradient boosting, and the learners of the papers it compares with; and (iii) report the composite and ranking measures of Section 7.6 together with the measures of the papers it wants to be compared with, even if it argues that another measure is more appropriate.

7.3 Benchmark Data and Tasks

The data decide what a comparison can show: a claim about severe imbalance needs datasets with severe imbalance, a claim of generality needs datasets that differ, and a comparison with earlier work needs the same datasets. This subsection describes the data the 71 studies use at the level of the evaluation, that is, where the data come from, how many datasets a study uses and why; the analysis of individual datasets and their difficulty is the subject of the Data section of this survey [TODO: reference to the Data section once its label is fixed].

Where the data come from. Two repositories supply most of the data (Table 6). Counted so that every study falls in exactly one group, 33 studies use UCI data but not KEEL data, 13 use KEEL data but not UCI data, 20 use both, and 5 use neither; in total 53 studies use UCI data and 33 use KEEL data. The five that use neither rely on private data (Kubat et al., 1998; Cieslak et al., 2006), the UCR time-series archive (Cao et al., 2013), public fraud and record-linkage data (Liu et al., 2020) or image benchmarks (Buda et al., 2018). The two repositories have different histories. UCI is used in every period by 69 to 86% of the studies. KEEL data first appear in 2013 and are then taken up quickly, by 38% of the studies of 2010 to 2015, 60% of those of 2016 to 2020 and 67% of those published since 2021. KEEL offers ready-made imbalanced versions of UCI datasets, named after the classes they oppose (for example ecoli-0-1_vs_5) and already split into five folds (Alcalá-Fdez et al., 2011), which is why the same dataset can be counted as UCI in one study and as KEEL in another, and why 20 studies use both. Synthetic data appear in 30 studies, often as small two-dimensional examples next to the real benchmarks, and private or application data in 11, mostly in the early periods (36% before 2010, 8% since 2021).

How many datasets. A study uses a median of 20 datasets (interquartile range 11 to 34, range 1 to 92), or 18 if only real datasets are counted (range 1 to 86). Benchmarks grew with KEEL: the median number of real datasets rose from 6 before 2010 to 14, 20 and 21.5 in the later periods, and the share of studies with fewer than ten real datasets fell from 64% to 4%. The same datasets recur. Ecoli is used in some form by 38 studies, yeast by 37, glass by 34, page-blocks by 30, new-thyroid and vehicle by 29 each and abalone by 25 (Table 7); most of these studies take them from KEEL (27 of the 38 ecoli studies, for example), and their use grew after 2010, when KEEL made imbalanced versions of them available. A few classic datasets belong to the UCI era and are now rarely used; satimage is the clearest case, with 8 of its 16 studies published before 2010.

Why these data. Fifty-eight of the 71 studies give a reason for their choice of data, and the share grew from 71% before 2010 to 88% since 2021 (Table 8). The reasons are mostly about coverage: a range of imbalance ratios, sizes or dimensions (30 studies), a target level of imbalance (19), the set-up of a named earlier study (18), and difficulty factors such as overlap, noise or small disjuncts (17). Fewer give comparability with published results (14), public availability (12) or an application domain (11) as the reason.

Table 6: Benchmark Data by Period

Studies (number)	All (71)	–2009 (14)	–2015 (13)	–2020 (20)	–2026 (24)
UCI only	33	12	7	6	8
KEEL only	13	0	3	4	6
Both UCI and KEEL	20	0	2	8	10
Neither	5	2	1	2	0
Total	71	14	13	20	24
Synthetic data	30	6	7	8	9
Private or application data	11	5	3	1	2
Image data	4	0	0	4	0
Fewer than 10 real datasets	19	9	4	5	1
States why the data were chosen	58	10	10	17	21
Median datasets per study (all)	20	6.5	20	20	23
Median real datasets per study	18	6	14	20	21.5

The first four rows partition the 71 studies, so they add up to 71: 53 use UCI data (33 only UCI, 20 both) and 33 use KEEL data (13 only KEEL, 20 both). The five that use neither use private data (2), the UCR time-series archive (1), public fraud and record-linkage data (1) and image benchmarks (1). All datasets includes synthetic ones.

Can the data be reproduced? In 53 studies all data are public, in 11 some are not, and 7 do not say. Public data are not enough, however, because most studies change them before use: 52 studies modify their data, for example by merging classes into a binary problem, subsampling a class or injecting noise, and only 10 of them release the modified versions. For the other 42 a reader knows the source but not the data on which the reported numbers were computed.

7.3.1 Tasks Other than Classification

Every one of the 71 studies evaluates resampling through a classification task. The studies differ in the number of classes (39 binary, 19 multi-class, 13 both) and in the type of data (65 tabular, 3 mixed, 2 image and 1 time series), but not in the task. Table 9 lists the studies whose data or output is not a plain table, and shows that even they come back to a classifier.

Two studies step partly outside classification, and it is worth seeing exactly how. CTGAN is a generator of synthetic tables rather than a resampler (Xu et al., 2019). Its main evaluation, which it calls machine learning efficacy, fits the generator on the training split of each real dataset, draws a synthetic table of the same size, trains several models on the synthetic table (a decision tree, a linear SVM, a multilayer perceptron, AdaBoost and logistic regression, and linear regression for the regression set) and scores them on the real test split, by F1 for binary labels, macro-F1 for several classes, accuracy for the MNIST-based sets and R^2 for its single regression dataset; the result is compared with the same models trained on the real training data. Its second evaluation needs no learner: on seven simulated datasets whose generating model is known (Gaussian mixtures and Bayesian networks), it computes the likelihood of the synthetic rows under the true model (L_{syn}) and the likelihood of real test rows under a model refitted on the synthetic rows (L_{test}), which falls when the generator covers only part of the distribution. GAMO generates minority-class images (Mullick et al., 2019). It trains a classification network on an imbalanced training set enlarged with the generated images and scores it on a balanced test set by average class accuracy and G-mean; separately, it computes the Fréchet inception distance between real and generated images of each class to compare the quality of its images with those of a conditional GAN. In both cases the evaluation that does not use a classifier either needs a known generating model (CTGAN's likelihoods) or measures image quality rather than the usefulness of the data (GAMO's distance).

Table 7: Dataset Families Used Most Often

Family	Studies	From KEEL	First year	–2009	–2015	–2020	–2026
ecoli	38	27	2010	0	9	12	17
yeast	37	26	2010	0	8	11	18
glass	34	26	2004	1	7	10	16
page-blocks	30	22	2007	1	6	10	13
new-thyroid	29	22	2004	1	5	10	13
vehicle	29	16	2004	3	4	12	10
abalone	25	12	2006	3	6	6	10
haberman	22	11	2004	4	5	8	5
pima	22	11	2002	6	4	6	6
cleveland	20	16	2010	0	5	5	10
vowel	19	12	2006	3	4	6	6
balance	16	10	2006	2	3	5	6
car	16	9	2006	2	3	4	7
satimage	16	2	2002	8	5	1	2
dermatology	15	13	2013	0	2	6	7
segment	15	7	2006	2	5	3	5
wine	15	10	2010	0	3	6	6
contraceptive	13	10	2013	0	1	5	7
hayes-roth	12	10	2013	0	1	5	6
led	12	11	2013	0	3	6	3
thyroid	11	11	2013	0	2	3	6
winequality-red	10	10	2016	0	0	3	7
zoo	10	8	2011	0	2	2	6
breast-tissue	9	6	2014	0	1	5	3
flare	9	8	2013	0	1	5	3

A study counts once for a family if it uses any dataset derived from it (for example ecoli0 vs 1 or ecoli-0-1-4-7 vs 2-3-5-6). From KEEL: studies that take the family from KEEL.

No study in the corpus evaluates regression as its main task, or clustering, anomaly detection, multi-label learning or data streams.

Why is everything classification? There are three reasons, and they explain also why direct measurement of the resampled data is rare (Section 7.7). First, the method is defined by class labels. SMOTE creates a new example between a minority example and one of its nearest neighbours of the same class, so it needs to know which examples form the minority class, and the problem it was designed for is a classifier that neglects that class. Success is therefore defined as a better classifier, and the evaluation goes back to a classifier. Second, the shared benchmarks are classification benchmarks: the imbalanced collections of KEEL and UCI are datasets with a designated minority class, and this literature has no shared benchmark for imbalanced regression, clustering or anomaly detection. Third, judging the resampled data without a classifier needs a reference that is usually missing. The true distribution of the minority class is unknown, the held-out minority examples needed to compare against are scarce by definition, and there is no agreed measure; a classifier score avoids all three problems, which is why even CTGAN and GAMO report one.

This reflects the corpus rather than a limit of the method. The fifteen-year review lists 22 SMOTE-based methods for other learning paradigms (Fernández et al., 2018): data streams, time series, semi-supervised and active learning, multi-instance and multi-label learning, regression, ordinal regression, text, image retrieval, imprecise data and high-dimensional data. These methods redefine what the minority is. SMOTER, for example, treats extreme values of a continuous target as rare cases, chosen by a user-defined threshold, and generates the target of a synthetic case as a weighted average of its two seeds (Torgo et al., 2013). Each paradigm also has its own measures,

Table 8: Reasons Given for Evaluation Choices

Choice	Reason given	Reasons given (studies)	Examples
Benchmark data	58 of 71	range of IR, size or dimension (30); a target imbalance level (19); set-up of a named earlier study (18); overlap, noise or small disjuncts (17); comparability with published results (14); public availability (12); an application domain (11)	(Barua et al., 2014; Saez et al., 2015; Koziarski et al., 2020)
Base learners	49 of 71	widely used or well known (22); different learning paradigms (15); set-up of a named earlier study (11); simple, fast or interpretable (10); same learner for all methods (8); sensitive to imbalance (4); matches how the method works (4)	(Napierala et al., 2010; Liu et al., 2023; Yan et al., 2025b)
Measures	70 of 71	standard in the literature (40); accuracy misleads under imbalance (39); focus on the minority class (25); independent of the threshold (18); suits several classes (18); costs of an application (7)	(Kubat and Matwin, 1997; Sun et al., 2015; Yadav and Vijay, 2025)
Measure used to select settings	33 of 54	G-mean (15); AUC (12); F1 (8); AvAcc (4); sensitivity (recall) (4); MAUC (3)	(Kubat et al., 1998; Cieslak and Chawla, 2008; Das et al., 2015)

A study can give several reasons. Selection criterion: studies that select a parameter value, variant or resampling amount.

such as a prequential AUC for streams with concept drift (Brzezinski and Stefanowski, 2017). Only one of these methods, INOS for time series, is in our corpus, and it is evaluated as a classifier of time series (Cao et al., 2013). For the reader, this sets the scope of this section: every statement about evaluation here holds for classification and should not be carried over to other paradigms without separate evidence.

What this means for a new study. Choose data that test the claim: public benchmarks that cover the range of imbalance ratios, sizes and difficulty the method is meant for, the benchmarks of the closest earlier studies, and, for a claim of generality, high-dimensional data, where SMOTE has been found to have hardly any effect on most classifiers (Blagus and Lusa, 2013). Give the reason for the choice in terms of what the data test rather than their popularity, keep multi-class data multi-class if the claim is about several classes, and release every modified dataset together with its folds.

7.4 Base Learners

How many and which. Almost half of the studies use one or two learner families. Across the 71 studies, 25 use one learner family, 9 use two, 12 three, 16 four and 9 five or more; the median is three. The share of single-learner studies has not fallen: it is 9 of the 24 studies since 2021. The most used families over the whole period are C4.5 (33 studies), SVM and k-NN (28 each), CART and the multilayer perceptron (17 each), naive Bayes (14) and logistic regression (13), with random forests (12) and gradient boosting (8) rising in the last two periods. Section 7.2 explains how the mix changed over time; Table 10 shows who uses each family, which toolkit those studies name, whether they release code, use KEEL data or address several classes, and how often the family is the only learner of a study.

A single learner limits what a study can conclude. A resampler changes the class proportions of the training data, and how much that helps depends on the learner: k-NN uses the same neigh-

Table 9: Studies Whose Data or Output Is Not a Plain Table, and How They Evaluate

Study	What is generated or resampled	How it is evaluated
CTGAN (2019) (Xu et al., 2019)	Generator of synthetic tables; 8 real datasets, one of them a regression task (news), and 7 simulated data sets with a known generating model	Train on synthetic, test on real: each of several learners is trained on the synthetic table and scored on the real test split (F1, macro-F1 or accuracy; R^2 for the regression set). On simulated data, likelihood of the synthetic rows under the true model (L_{syn}) and of real test rows under a model refitted on synthetic rows (L_{test})
GAMO (2019) (Mullick et al., 2019)	Generative oversampling of minority images; 7 image benchmarks made imbalanced by subsampling, plus tabular sets in a supplement	Classification of a fixed, balanced test set (average class accuracy and G-mean); separately, the Fréchet inception distance between real and generated images of each class
INOS (2013) (Cao et al., 2013)	Oversampling of time series; 7 UCR data sets split into 48 binary tasks	Classification by an SVM: F1, G-mean, precision, recall and TNR on random hold-out splits
Buda et al. (2018) (Buda et al., 2018)	Random oversampling for convolutional networks; MNIST, CIFAR-10 and ImageNet made imbalanced	Classification of a balanced test set: multi-class ROC-AUC and accuracy

Table 10: Who Uses Each Learner Family

Family	Studies	Python	Weka	Code	KEEL data	Multi-class	Alone	Datasets
C4.5 or C5.0	33	2	11	8	15	16	9	20
CART (decision tree)	17	10	1	13	7	11	2	21
Rule learner	9	0	4	1	2	4	2	19
k-NN	28	11	6	17	16	13	0	23
Support vector machine	28	10	5	10	15	10	5	23.5
Logistic regression	13	6	0	9	4	5	0	19
Random forest	12	6	2	7	6	4	1	26.5
Gradient boosting	8	7	0	6	4	3	1	34.5
Naive Bayes	14	6	4	6	8	8	0	20
Multilayer perceptron	17	3	3	7	5	10	3	20

All counts are studies among the 71. Python, Weka: the study names the toolkit (a lower bound). Code: the study releases code. Alone: the family is the only learner of the study. Datasets: median number of datasets of the users.

bourhoods that SMOTE interpolates in, decision trees are sensitive to pruning that removes the rules for small classes, and strong ensembles already compensate for part of the imbalance. With one learner a reader cannot tell whether a method is better in general or only fits that learner. The 25 single-learner studies use C4.5 (9), SVM (5), the multilayer perceptron (3), and other families once or twice each.

Why these learners. Forty-nine of the 71 studies say why they chose their learners (Table 8). The most frequent reason is that the learners are widely used or well known, given by 22 studies and, in 8 of them, the only reason. The studies do write this, often in so many words: logistic regression is chosen "as it is a commonly used classification algorithm" (Liu and Hsieh, 2020), three learners are adopted as "three of the best-known classification algorithms" (Zheng et al., 2022), and the very first study in the corpus selected its learners "because of their widespread use and well-known performance" (Kubat and Matwin, 1997). Other reasons are that the learners cover different learning paradigms (15 studies), that a named earlier study used them (11), that they are

simple, fast or interpretable (10), and that the same learner is used for every method so that the comparison is fair (8).

Popularity is a weak reason. It explains why a reader will recognise the learner, not what the learner tests, and it would justify the same learners for any method on any data. What a reader needs to know is which property of a learner makes it a meaningful test of a resampler: whether its decision rule depends on the class proportions it sees, which is what resampling changes; whether it produces scores, which ranking measures such as ROC-AUC need; whether it shares the neighbourhood logic of SMOTE, as k-NN does, or is robust to imbalance, as random forests and boosting partly are; and whether it is the learner of the work being compared with. Only 4 studies choose a learner because it is sensitive to imbalance (Napierala et al., 2010; Saez et al., 2015; Yan et al., 2025b; Tao et al., 2025), and 4 because it matches how the method works, for example a boosting method that needs an unstable base learner (Diez-Pastor et al., 2015). Following an earlier study is a sound reason when the study is named, because it makes the results comparable; 11 studies name one, as when the MC-CCR line keeps the learners of its first two papers (Liu et al., 2023). A new study should give, for each learner, the property it tests or the study it allows comparison with.

What this means for a new study. Use at least four learners from different families: a decision tree (CART or C4.5), k-NN, a linear or margin-based model (logistic regression or an SVM) and a strong ensemble (random forest or gradient boosting), configured identically for every resampler, and report results per learner rather than only an average over learners. Name the library and version of each, since "decision tree" means C4.5 in Weka and CART in scikit-learn, and say which score feeds the ranking measures (a probability or a decision function).

7.5 Competitors and Baselines

A comparison is only as convincing as what the proposal is compared with. A study compares its proposal with a median of 7 other methods (range 2 to 30). Plain SMOTE is one of them in 50 of the 71 studies, and its share rose from 50% of the studies before 2010 to 79% since 2021, so SMOTE remains the standard reference. A baseline without any resampling appears in 52 studies, but here the trend runs the other way: it was included by 86% of the studies before 2010 and by only 54% of those published since 2021 (Table 11). Random oversampling is included by 27 studies and Borderline-SMOTE by 27, ADASYN by 19 and random undersampling by 18, the last falling from half of the early studies to 8% of the recent ones. Recent studies compare with recent methods: MDO is a competitor in 38% of the studies since 2021 and MC-CCR in the same share, almost all of them in the multi-class line of work.

Baselines that do not resample. The most informative baselines are those that address imbalance without creating data. A cost-sensitive learner or instance weighting is a competitor in only 5 studies (Fernandez et al., 2013; Yang et al., 2018; Yao and Lin, 2021; Xie et al., 2022; Yadav and Vijay, 2025), three of them published after the fifteen-year review restated the finding that preprocessing and cost-sensitive learning are equally good ways to address imbalance (López et al., 2012; Fernández et al., 2018). Plain class weighting of the learner appears in one study (Yadav and Vijay, 2025). Threshold moving, which changes only the decision threshold of a trained learner, appears in three: as threshold-moving neural networks and probability estimation trees in a study of multi-class cost-sensitive learning (Zhou and Liu, 2010), as a threshold-moving neural network in DSRBF (Fernandez-Navarro et al., 2011) and as thresholding by the class priors in a study of convolutional networks (Buda et al., 2018). Cost-sensitive ensembles are somewhat more common:

Table 11: Competitors and Baselines: Studies That Include Each, by Period

Competitor	Studies (71)	–2009 (14)	–2015 (13)	–2020 (20)	–2026 (24)
SMOTE	50	50%	77%	70%	79%
No resampling	52	86%	85%	80%	54%
Random oversampling (ROS)	27	36%	31%	60%	25%
Borderline-SMOTE	27	7%	15%	50%	58%
ADASYN	19	0%	23%	35%	38%
Safe-Level-SMOTE	8	0%	8%	15%	17%
Random undersampling (RUS)	18	50%	31%	25%	8%
MWMOTE	8	0%	0%	15%	21%
SMOTE+ENN or SMOTE+Tomek	12	0%	23%	10%	29%
MDO	13	0%	0%	20%	38%
k-means SMOTE	9	0%	0%	10%	29%
RBO or MC-RBO	7	0%	0%	0%	29%
MC-CCR or CCR	9	0%	0%	0%	38%
SMOTEBoost	9	7%	38%	5%	8%
RUSBoost	7	0%	23%	10%	8%
EasyEnsemble	5	0%	15%	5%	8%

Period columns give the share of the studies of that period. SMOTE and no resampling are coded directly; the other rows are matched on the names in each study's list of competitors.

AdaCost is a competitor of SMOTEBoost and RAMOBoost (Chawla et al., 2003; Chen et al., 2010) and MetaCost of SplitBal (Sun et al., 2015).

These baselines matter because of the way resampling acts. Section 7.6.1 shows that balancing the training data mainly moves the operating point of a classifier towards the minority class, and that G-mean and other threshold measures reward that movement. Class weighting, threshold moving and random oversampling move the operating point too, at almost no cost. A gain of a new resampler is therefore evidence of something more than a shifted threshold only if it exceeds what those simple baselines achieve. The one study that includes class weighting, run under a protocol it registered in advance, found its new oversampler no better than class weighting, random oversampling or no resampling across four learners (Yadav and Vijay, 2025).

What this means for a new study. Include four simple baselines, namely no resampling, class weighting (or threshold moving) of the learner, random oversampling and plain SMOTE, and the two to four strongest recent competitors, preferring those with public code. Run every method on the same folds, with maintained implementations such as imbalanced-learn (Lemaître et al., 2017) or a stated version of the authors' code, and check any re-implementation against at least one result of the original paper. Do not copy numbers from other papers, whose folds, settings and sometimes definitions of the measures differ.

7.6 Performance Measures

The measure turns the predictions of a classifier into the number a paper reports, and it is the choice that most often decides which method comes out ahead. This subsection defines the measures in use, counts them, and asks which is used most and why, and how the two most used composites differ.

Notation. Consider a binary problem in which the minority class is the positive class. A classifier that outputs a score $s(x)$ predicts the positive class when $s(x) \geq t$ for a threshold t. On a test

set, TP and FN count the positive examples predicted positive and negative, FP and TN count the negative examples predicted positive and negative, $\pi = (\mathrm{TP}+\mathrm{FN})/n$ is the share of positives and $\mathrm{IR} = (1-\pi)/\pi$ is the imbalance ratio. The measures used in the corpus are built from four rates:

$$\mathrm{TPR} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \qquad \mathrm{TNR} = \frac{\mathrm{TN}}{\mathrm{TN}+\mathrm{FP}}, \qquad \mathrm{FPR} = 1-\mathrm{TNR}, \qquad \mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}. \quad (1)$$

TPR is also called recall or sensitivity, and TNR specificity. TPR and TNR are computed within one class each and do not depend on π; precision mixes the two classes and does. The threshold composites combine these rates at one threshold:

$$\text{G-mean} = \sqrt{\mathrm{TPR}\cdot\mathrm{TNR}}, \qquad F_1 = \frac{2\cdot\mathrm{Precision}\cdot\mathrm{TPR}}{\mathrm{Precision}+\mathrm{TPR}} = \frac{2\,\mathrm{TP}}{2\,\mathrm{TP}+\mathrm{FP}+\mathrm{FN}}, \quad (2)$$

$$\mathrm{BA} = \frac{\mathrm{TPR}+\mathrm{TNR}}{2}, \quad (3)$$

$$\mathrm{MCC} = \frac{\mathrm{TP}\cdot\mathrm{TN}-\mathrm{FP}\cdot\mathrm{FN}}{\sqrt{(\mathrm{TP}+\mathrm{FP})(\mathrm{TP}+\mathrm{FN})(\mathrm{TN}+\mathrm{FP})(\mathrm{TN}+\mathrm{FN})}}. \quad (4)$$

The general

$$F_\beta = \frac{(1+\beta^2)\,\mathrm{TP}}{(1+\beta^2)\,\mathrm{TP}+\beta^2\,\mathrm{FN}+\mathrm{FP}}$$

weights recall β times as much as precision. The ranking measures do not fix a threshold. ROC-AUC is the area under the curve of TPR against FPR as t varies, which equals the probability that a randomly drawn positive example receives a higher score than a randomly drawn negative one (Bradley, 1997); PR-AUC is the area under the curve of precision against recall (Saito and Rehmsmeier, 2015). For K classes with per-class recall r_k, the multi-class G-mean is $\mathrm{MG} = (\prod_{k=1}^{K} r_k)^{1/K}$, the average accuracy used in this literature is $\mathrm{AvAcc} = \frac{1}{K}\sum_k r_k$, and MAUC averages the pairwise AUC of every pair of classes (Hand and Till, 2001). CBA divides each diagonal entry of the confusion matrix by the larger of its row and column sums and averages over classes, and CEN measures the entropy of the misclassifications. Table 12 defines every measure found in the corpus. One name needs care: in the literature on data complexity, F1 also denotes Fisher's discriminant ratio, the most used measure of class overlap (Ho and Basu, 2002; Fernández et al., 2018), so a study that reports both should name them differently.

How many measures, and how they group. Across the 71 studies we found 27 distinct scalar measures computed from the predictions of a downstream classifier. They fall into five families and one further measure (Table 13): overall accuracy or error (1 measure, 13 studies), single-class rates (4 measures, 32 studies), threshold composites (7 measures, 56 studies), threshold-free ranking measures (4 measures, 44 studies) and multi-class aggregates (10 measures, 21 studies), plus misclassification cost (2 studies). Counting F1 and an F-measure whose β is not stated as one measure gives 26. Every empirical study reports at least one. A study reports a median of three: 5 studies report one measure, 18 two, 19 three, 12 four, 10 five and 7 six or more, so 23 studies rely on one or two.

The families answer different questions, and choosing among them is the most consequential decision of an evaluation. A composite such as G-mean, balanced accuracy or MCC asks whether the classifier, at the threshold it actually uses, treats both classes well. A ranking measure such as ROC-AUC or PR-AUC asks whether the classifier orders minority examples above majority ones, whatever threshold is later chosen. Resampling acts mainly on the first question, as Section 7.6.1 shows, so a study that reports only composites cannot tell whether a resampler improved the

Table 12: The 27 Scalar Measures Found in the Corpus

Measure	Family	Meaning	Definition	Studies
Accuracy or error	Overall	share of all predictions that are correct (error: 1 − accuracy)	$(TP + TN)/n$	13
Recall (TPR, sensitivity)	Single class	share of minority examples found	$TP/(TP + FN)$	29
Specificity (TNR)	Single class	share of majority examples rejected	$TN/(TN + FP)$	5
Precision	Single class	share of minority predictions that are correct; depends on the class prior	$TP/(TP + FP)$	20
FPR	Single class	false positive rate	$1 - \text{TNR}$	1
G-mean	Composite	geometric mean of the two class rates	$\sqrt{\text{TPR} \cdot \text{TNR}}$	41
F1	Composite	harmonic mean of precision and recall	$2PR/(P + R)$	24
F-measure (F_β)	Composite	weights recall β times as much as precision	$(1 + \beta^2)PR/(\beta^2 P + R)$	11
Balanced accuracy	Composite	arithmetic mean of the two class rates	$(\text{TPR} + \text{TNR})/2$	3
MCC	Composite	correlation between predicted and true labels; uses all four cells	Equation 4	4
Kappa	Composite	agreement corrected for chance	$(p_o - p_e)/(1 - p_e)$	1
G-score	Composite	geometric mean of precision and recall (not G-mean)	$\sqrt{P \cdot R}$	1
ROC-AUC	Ranking	probability that a minority example is scored above a majority one	area under TPR against FPR	33
PR-AUC (AUPRC)	Ranking	area under precision against recall; depends on the prior	area under P against R	5
MAUC	Ranking	average AUC over all pairs of classes (Hand and Till, 2001)	$\frac{2}{K(K-1)} \sum_{i<j} \hat{A}(i, j)$	12
AUCm	Ranking	other multi-class AUC, for example one-vs-rest average	mean of per-class AUC	2
Per-class recall	Multi-class	recall of each class reported separately	r_k	4
Macro-F1	Multi-class	F1 averaged over classes with equal weight	$\frac{1}{K} \sum_k F1_k$	7
AvAcc	Multi-class	mean per-class recall (some studies use mean one-vs-rest accuracy, which true negatives inflate)	$\frac{1}{K} \sum_k r_k$	7
Weighted F1	Multi-class	F1 averaged with class-size weights; dominated by large classes	$\sum_k w_k F1_k$	1
Macro-precision	Multi-class	precision averaged over classes	$\frac{1}{K} \sum_k P_k$	4
Pmin	Multi-class	smallest per-class precision	$\min_k P_k$	1
CBA	Multi-class	class balance accuracy: each diagonal cell over the larger of its row and column sums, averaged	$\frac{1}{K} \sum_k c_{kk} / \max(c_{k\cdot}, c_{\cdot k})$	6
CEN	Multi-class	confusion entropy, the spread of the errors (lower is better)	entropy of the confusion matrix	5
Macro-recall	Multi-class	recall averaged over classes (equals AvAcc when AvAcc is defined correctly)	$\frac{1}{K} \sum_k r_k$	1
Minimum sensitivity	Multi-class	smallest per-class recall	$\min_k r_k$	2
Misclassification cost	Cost	total cost of the errors under a cost matrix	$\sum C(i, j)\, n_{ij}$	2

TP, FN, FP, TN: cells of the binary confusion matrix with the minority class as positive; P precision, R recall, r_k and P_k recall and precision of class k of K, c_{ij} cells of the multi-class confusion matrix. ROC curves (8 studies), confusion matrices (4), running time (14) and the fidelity of synthetic data (3) are also reported but are not scalar measures of classification.

Table 13: Families of Measures Computed from a Downstream Classifier

Family	Measures in the corpus	Distinct	Studies	Question it answers	Main limitation
Overall	Accuracy, error rate	1	13	How many predictions are correct?	Dominated by the majority class; a classifier that ignores the minority scores π_-
Single-class rates	Recall (TPR), specificity (TNR), precision, FPR	4	32	How well is one class handled?	One rate alone rewards predicting everything as minority
Threshold composites	G-mean, F1, Fβ, balanced accuracy, MCC, kappa, G-score	7	56	Does the classifier treat both classes well at its operating point?	Depend on the threshold; F ignores true negatives and depends on the prior
Threshold-free ranking	ROC-AUC, PR-AUC, MAUC, AUCm	4	44	Does the classifier rank minority examples above majority ones?	Need scores; ROC-AUC can look good under severe imbalance; PR-AUC depends on the prior
Multi-class aggregates	MG, AvAcc, macro-F1, CBA, CEN, Pmin, per-class recall and others	10	21	Is every class, including the smallest, handled well?	The averaging rule changes the value; it is often not stated
Cost	Misclassification cost	1	2	What do the errors cost?	Needs costs that benchmark data do not have

Distinct: measures of the family named in at least one study (27 in total). Studies: studies that report at least one measure of the family. ROC curves, confusion matrices, running time and fidelity of synthetic data are not scalar measures of classification and are not counted.

classifier or merely moved its threshold, and a study that reports only ranking measures may miss the effect a practitioner cares about. Reporting one measure of each kind separates the two. For multi-class problems the averaging rule is part of the measure and must be stated: MG is zero as soon as one class has zero recall, AvAcc is not, and macro-, micro- and weighted averages of F1 can order methods differently.

The primary measure. Fifty-six studies name the measure that decides their main conclusion. ROC-AUC is named most often (16 studies, and one more names the ROC curve), followed by an F-measure (8), G-mean (7), AvAcc (4), recall or a sensitivity-based measure (4), PR-AUC (2) and misclassification cost (2); 10 studies, most of them multi-class, name two or more measures jointly, such as MAUC and MG. The other 15 report several measures without saying which one decides.

Where the measures come from. When the studies justify or define their measures they cite a small set of sources (Table 14): Bradley's analysis of AUC (Bradley, 1997) and the survey of He and Garcia (11 studies each), Hand and Till's MAUC (Hand and Till, 2001) (9), and then work on ROC analysis and on the measures of Kubat and colleagues (Kubat and Matwin, 1997). The reasons they give are discussed with G-mean below (Table 16).

Definitions are often wrong or missing. In 14 of the 71 studies (20%) at least one measure is defined in a way that changes its meaning, or its definition cannot be recovered (Table 15). Two studies call the single-threshold quantity $(1 + \text{TPR} - \text{FPR})/2$, which is balanced accuracy, "AUC" (Datta et al., 2020; Xie et al., 2022); one defines G-mean as $\sqrt{\text{Recall} \cdot \text{Precision}}$ (Liu et al., 2020); two compute MAUC from hard labels instead of scores (Zheng et al., 2022; Yan et al., 2025b); two define AvAcc as the mean one-vs-rest accuracy, which is inflated by true negatives (Ma et al., 2024; Tao et al., 2025); and others leave the multi-class averaging unstated (Agrawal et al., 2015; Palli et al., 2022), or, in one case, define F1 as the harmonic mean of TPR and TNR (Palli et al.,

Table 14: Sources Cited When Measures Are Chosen or Defined

Source	Studies citing it
Bradley 1997	11
He and Garcia 2009	11
Hand and Till 2001	9
Swets 1988	4
Provost and Fawcett 2001	4
Provost and Fawcett 1997	4
Fawcett 2003	4
Maloof 2003	4
Kubat and Matwin 1997	4
Fawcett 2006	4
Ferri et al. 2009	4
Wang and Yao 2012	4
Branco et al. 2017	4
Kubat et al. 1997	3

2022). The single-threshold "AUC" deserves most attention, because some tools report it under that name, so a reader comparing AUC values across papers may be comparing two different quantities. These problems also go with fewer citations. Studies with a definition problem have a median of 25 citations, against 218.5 for the others; they are more recent (median year 2022 against 2017.5), but the gap remains within each period: 105 against 292.5 citations among the studies of 2010 to 2015, 59 against 260.5 among those of 2016 to 2020, and 18 against 41 among those published since 2021.

What this means for a new study. Define every measure with a formula, state which class is positive, state the threshold (the default of the learner, a tuned threshold, or none for ranking measures) and, for multi-class problems, the averaging rule. Report at least one threshold composite and one ranking measure; for a highly imbalanced problem, add PR-AUC to ROC-AUC. Declare the primary measure before running the experiments, and give the reason from the application rather than from the results.

7.6.1 Why G-mean Became the Most Reported Measure

G-mean is reported by 41 of the 71 studies (58%), more than any other measure, followed by F1 or F-measure (35), ROC-AUC (33), recall (29) and precision (20). Its share rose in every period, from 29% before 2010 to 54%, 65% and finally 71% of the studies published since 2021, and it is also the most used measure in multi-class studies, where it appears as MG (11 of 19). It is the only one of the main measures that rose steadily: over the same four periods ROC-AUC moved from 36% to 62%, 35% and 54%, F1 from 43% to 31%, 35% and 29%, and recall reported on its own fell from 71% to 38%, 45% and 21%. Two qualifications matter. When a paper names a single primary measure, that measure is more often ROC-AUC (16 studies) than G-mean (7), and among the 24 binary-only SMOTE-family methods ROC-AUC is still slightly ahead (18 against 14). G-mean is therefore the measure most often reported, not the one most often trusted as the final word.

Why G-mean? The answer the studies give is mostly popularity (Table 16). Of the 41 G-mean users, 24 say that their measures are standard in the literature and 21 that accuracy misleads under imbalance, but these reasons apply to any measure other than accuracy, and the studies that do not use G-mean give them almost as often (16 and 18 of 30). Only about six studies give a reason specific to G-mean: that it addresses the inadequacy of accuracy for this kind of problem (Kubat et al., 1998), that it suits rule and tree learners that output labels rather than scores (Napierala

Table 15: Studies With a Problem in the Definition of a Measure

Study	Problem
SCUT (2015) (Agrawal et al., 2015)	multi-class averaging of G-mean, F-measure and AUC not defined
CGMOS (2016) (Zhang et al., 2016)	'G-score' = sqrt(Precision x Recall), reported for both classes as positive
AMDO (2018) (Yang et al., 2018)	Pavg described as precision but equated with average accuracy
DLexiBoost (2020) (Datta et al., 2020)	'AUC' computed from crisp predictions as (1+TPR-FPR)/2 (= balanced accuracy)
SPE (2020) (Liu et al., 2020)	G-mean defined as sqrt(Recall x Precision) instead of sqrt(TPR x TNR)
EMDO (2021) (Yao and Lin, 2021)	per-class precision called 'accuracy' (Pmin/Pavg)
CBHSID (2022) (Palli et al., 2022)	multi-class averaging of Recall/F1 not stated; F1 misdefined
GDO (2022) (Xie et al., 2022)	'AUC' defined as (Sens+Spec)/2, i.e. single-threshold balanced accuracy
SHSampler (2022) (Zheng et al., 2022)	MAUC computed from predicted labels instead of scores
SOS (2022) (Kim et al., 2022)	weighted-F1 (majority-dominated) as the sole main metric
MC-MBRC (2024) (Ma et al., 2024)	AvAcc defined as mean one-vs-rest accuracy, inflated by true negatives
GDHS (2025) (Yan et al., 2025b)	MAUC computed from predicted labels instead of scores
MC-NCLWO (2025) (Tao et al., 2025)	AvAcc defined as mean one-vs-rest accuracy, inflated by true negatives
Loyalty-SMOTE (2026) (Hu et al., 2026)	implausible AUC values (e.g. 0) and averages inconsistent with tables

et al., 2010), that it rewards a gain in sensitivity only if both classes keep good performance (Das et al., 2015), that it is high only when every class is handled well (Datta et al., 2020), that it balances the two class accuracies (Maldonado et al., 2022), and that it ensures no class is overlooked (Wang and Awang, 2024). Popularity, however, is a consequence that needs its own cause. The cause is the way resampling acts on a classifier, and it can be stated as a chain:

1. Resampling changes the class proportions of the training data, usually to 1:1.
2. A classifier trained on balanced data predicts the minority class more often. Its main effect is to move the operating point: TPR goes up and TNR goes down. How well the classifier separates the two classes, which ROC-AUC measures, changes much less.
3. G-mean measures precisely this trade-off. It is maximal when TPR and TNR are both high and equal, and it is zero when either class is ignored.
4. For two classes whose scores follow normal distributions with equal variance, the threshold that maximises G-mean lies midway between the two class means, which is exactly the threshold that a classifier trained on balanced data adopts, whatever the imbalance ratio. Balanced resampling moves the classifier to the point that G-mean rewards most.
5. A proposed resampler therefore shows its clearest gain under G-mean, and later studies report G-mean to be comparable with it.

Our own illustrative runs, described in Appendix A, show the size of the effect (Figure 4). Over 110 KEEL datasets and five learners, SMOTE beat no resampling in 427 of the 550 dataset-learner pairs (78%) when measured by G-mean, with a median gain of 0.064. Measured by F1 it won 58% of the pairs (median gain 0.014), by ROC-AUC 46% (median gain 0.000) and by PR-AUC only 32% (median change −0.001). This is what the chain predicts: resampling helps the measures that depend on the threshold and does little for the measures that do not.

Table 16: Reasons Stated for the Choice of Measures

Reason given	G-mean users ($n = 41$)	Other studies ($n = 30$)
Standard in the literature	24 (59%)	16 (53%)
Accuracy misleads under imbalance	21 (51%)	18 (60%)
Focus on the minority class	13 (32%)	12 (40%)
Independent of the threshold	11 (27%)	7 (23%)
Suits several classes	11 (27%)	7 (23%)
Costs of an application	4 (10%)	3 (10%)
Other, specific to a measure	13 (32%)	11 (37%)

A study can give several reasons; 70 of the 71 studies give at least one.

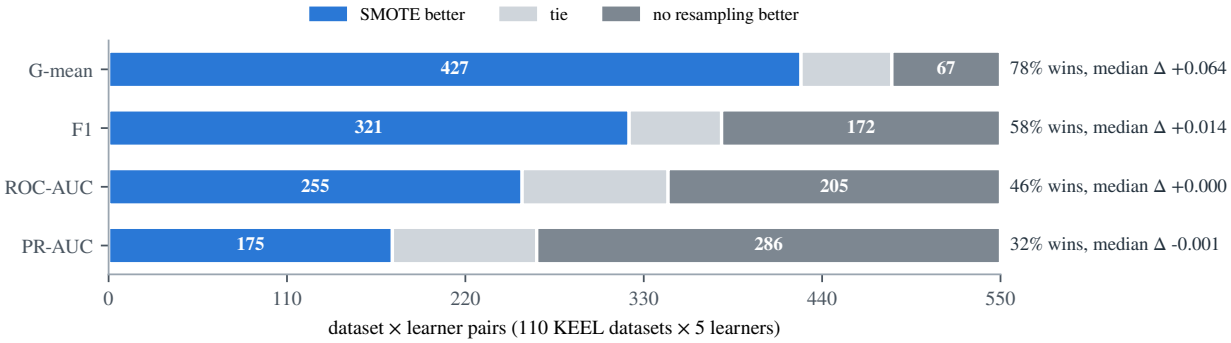


Figure 4: SMOTE Against No Resampling, by Measure (Illustrative Runs)

Three properties of G-mean then made it convenient for the way this literature evaluates, which explains why it spread further than other composites. It does not depend on the class prior, because TPR and TNR are computed within each class, so the values for a dataset with IR 1.8 and one with IR 130 can be ranked together; this matters when a study averages ranks over 30 to 100 KEEL datasets, and 23 of the 41 G-mean users use KEEL data. It needs only predicted labels, so it works with C4.5, rule learners and k-NN, which produce poor scores. And it extends to several classes without a choice of averaging rule (MG). Its lineage helped as well: the first two studies in the corpus that evaluate on imbalanced data, OSS and SHRINK (Kubat and Matwin, 1997; Kubat et al., 1998), used it as their main measure, and eight studies cite work by Kubat and colleagues when they explain their measures.

The same mechanism is a warning. Because G-mean rewards any movement of the operating point towards balance, it rewards methods that do nothing but move the threshold, which is why the baselines of Section 7.5 matter. G-mean also ignores precision: on a dataset with a large imbalance ratio, a classifier can reach a high G-mean while most of its positive predictions are wrong. Finally, 13 studies use G-mean to choose their settings on the same data on which they report it, which inflates exactly the number the reader sees (Section 7.8).

What this means for a new study. G-mean is a reasonable measure for a resampler, but it should never be the only one. Report it together with a ranking measure, include class weighting or threshold moving and random oversampling as baselines, and if the method is tuned, tune it on data that are not used to report G-mean.

7.6.2 G-mean Versus F1

G-mean and F1 are the two most reported composites and are often reported side by side, so a reader should know when they disagree. Both contain recall. They differ in the second component (Equation 2). G-mean asks what share of each class is classified correctly and takes the geometric mean of the two shares; each share is computed within its own class, so the size of the classes does not enter. F1 combines recall with precision, the share of positive predictions that are correct, and does not use the true negatives at all. Because precision mixes the two classes, it depends on the prior:

$$\text{Precision} = \frac{\text{TPR} \cdot \pi}{\text{TPR} \cdot \pi + \text{FPR} \cdot (1 - \pi)}, \qquad \pi = \frac{1}{1 + \text{IR}}. \tag{5}$$

When IR is large, π is small, and a false positive rate that is small as a percentage of the majority class becomes a large number compared with the minority class, so precision falls. TNR contains no π, so G-mean does not change. Averaging F1 over benchmarks whose imbalance ratios range from 1.8 to 130, as KEEL benchmarks do, therefore mixes the quality of a method with the difficulty implied by each dataset's IR. G-mean does not have this problem, which is one more reason it suits multi-dataset comparisons.

The two measures also prefer different thresholds. G-mean is maximised at the balanced threshold of Section 7.6.1. F1 penalises the false positives that a balanced classifier makes, so its best threshold lies between the balanced threshold and the threshold that maximises accuracy on the original data, and it moves towards the latter as IR grows. Balanced resampling therefore helps F1 less than G-mean, and when the classes are well separated and IR is large it can lower F1. The choice between the two measures alone can decide which method a paper declares the winner.

When to use which. Use G-mean, or balanced accuracy, when errors on both classes matter, for example in diagnosis, where a missed patient and a false alarm both have a cost; when results are averaged over datasets with very different imbalance ratios; when there are several classes (as MG); and when the learner gives only labels. Use F1 when the positive class is the focus and the cost lies in positive predictions that turn out wrong, for example when each positive prediction triggers a manual check or in document retrieval, where the negative class is background that need not be reported, and when results are to be compared with work in retrieval or detection.

What this means for a new study. Report both, together with a ranking measure (ROC-AUC and, under heavy imbalance, PR-AUC), because the three answer different questions and can name different winners. If one of them is to be the primary measure, declare it before looking at the results and give the reason from the problem, not from the results, and do not average F1 over datasets with very different IR without also reporting ranks.

7.7 Evidence Beyond Classifier Scores

A resampler produces data, yet almost every study judges those data only through a classifier. Only 6 of the 71 studies (8%) examine the resampled data directly, and none of them uses such a check instead of a classifier; it is always additional evidence (Table 17). Two-dimensional plots of synthetic samples, usually on a toy dataset, are more common (33 studies) but show a picture rather than a number, and 23 studies report nothing but classifier scores.

The six direct checks are of four kinds. Three studies measure how far the synthetic data are from the real distribution: CTGAN computes the likelihood of synthetic rows under a known generating model (Xu et al., 2019), GAMO the Fréchet inception distance between real and generated

Table 17: Studies That Check the Resampled Data Directly

Study	What is checked	How
SMOTE-IPF (2015) (Saez et al., 2015)	Noise created by SMOTE	Share of synthetic, original minority and original majority examples removed by a noise filter after SMOTE, on synthetic, real and noise-injected data
SPE (2020) (Liu et al., 2020)	Hardness of the examples kept	Number of majority examples per hardness bin before and after under-sampling; resampled sets drawn on a 2-D checkerboard
CTGAN (2019) (Xu et al., 2019)	Fidelity to the true distribution	Likelihood of synthetic rows under the known generating model, and of real test rows under a model refitted on synthetic rows
GAMO (2019) (Mullick et al., 2019)	Distance between distributions (images)	Fréchet inception distance between real and generated images; a proof that the objective is a sum of Jensen–Shannon divergences
GLOS (2023) (Han et al., 2023)	Type of the minority examples	Counts of safe, borderline and small-disjunct minority examples as a parameter varies
CISO (2025) (Yadav and Vijay, 2025)	Distance to held-out minority data; per-sample guarantee	MMD^2 and the AUC of a classifier two-sample test against held-out minority examples; share of synthetic points that pass a certificate of not entering majority regions
Hassanat et al. (2022) (Hassanat et al., 2022)	Whether synthetic points belong to the minority	Hide 10, 25 or 50% of the majority class, oversample the rest, and count the synthetic points whose nearest original example (Hassanat distance) is a majority example; no classifier is trained

The last row is a position paper that is not among the 71 empirical studies.

images (Mullick et al., 2019), and CISO the maximum mean discrepancy and the AUC of a classifier trained to tell synthetic points from held-out minority points (Yadav and Vijay, 2025). Two count what kind of examples the method produces or keeps, which shows where on the decision boundary resampling adds or removes points: GLOS counts safe, borderline and small-disjunct minority examples (Han et al., 2023), and SPE counts majority examples by hardness (Liu et al., 2020). One counts noise: SMOTE-IPF measures how many synthetic examples a noise filter removes after SMOTE, that is, how many fall in majority regions (Saez et al., 2015). The fourth kind certifies single points: CISO checks that each synthetic point does not enter a majority region. Outside the empirical studies, a position paper proposes a check of the same kind that needs no classifier at all: hide part of the majority class, oversample the rest and count the synthetic points whose nearest original example is a hidden majority example (Hassanat et al., 2022). Five further studies inspect the trained model rather than the data, by the size of the tree (Chawla et al., 2002; Janicka et al., 2019), the number and length of the rules (Batista et al., 2004), the diversity of ensemble members (Diez-Pastor et al., 2015; Zhu et al., 2023).

Why are direct checks so rare? A direct check answers whether the synthetic data are good, not whether the classifier is better, and the second question is the one reviewers and users ask. There is also no common measure: the six studies use six different checks, most of which need either a known generating distribution or held-out minority examples, which are scarce by definition (Section 7.3.1). Yet direct checks catch failures that scores hide. A resampler that places synthetic points inside majority regions can still raise G-mean, because it moves the threshold (Section 7.6.1), while making the minority class less faithful. The fifteen-year review asked for exactly this kind of analysis: it named the relationship between the synthetic instances and the intrinsic characteristics of the data (small disjuncts, noise, overlap) as the main open question (Fernández et al., 2018). The response has been modest. Five of the 36 studies published from 2019 check the resampled data directly, against one of the 35 before, and only two of them count the safe, borderline or rare examples that the review's challenges refer to (Liu et al., 2020; Han et al., 2023). The six studies with a direct check have a median of 238.5 citations against 140.5 for the others, at a similar median year (2019.5

against 2018), but with six studies the comparison within periods points both ways and shows no clear pattern.

What this means for a new study. Add at least one cheap direct check, computed on held-out data. Two need no assumptions about the data: the share of synthetic points whose nearest original neighbours are mostly majority examples, in the spirit of Hassanat et al. (Hassanat et al., 2022), and a two-sample test (MMD or a classifier two-sample test) between synthetic points and minority examples held out from resampling, as in CISO (Yadav and Vijay, 2025). A third relates the method to the difficulty of the data: the shares of safe, borderline, rare and outlying minority examples before and after resampling (Napierała and Stefanowski, 2016). Report the checks per dataset next to the classifier scores, and use two-dimensional plots as illustrations only.

7.8 Validation Protocol and the Choice of Settings

The measures of the previous subsections are only as good as the experiment that produces them. Two steps decide whether a reported difference can be trusted before any test is run: how the data are split, and how the settings of each method are chosen. Table 19 shows both by period.

Splitting the data. Stratified k-fold cross-validation is the norm (Table 18): 33 studies repeat it, 16 run it once and 4 use 5×2 cross-validation, with $k = 10$ (23 studies) and $k = 5$ (20) as the usual choices. Eight studies use a single hold-out split, 6 repeated hold-out splits, and 3 deep or generative studies the fixed splits of their benchmarks (Buda et al., 2018; Xu et al., 2019; Mullick et al., 2019); one study does not describe its protocol (Datta et al., 2020). In total 43 of the 71 studies (61%) average over repeated partitions, a share that rose from 43% before 2010 to 71% since 2021; the rest report a single estimate per dataset, so their differences between methods include the noise of one particular split. The choice of k changed with the data. Ten folds were the rule before 2016 (57% and 62% of the studies), and five folds are the rule since 2021 (54%, against 8% for ten). The choice of $k = 5$ is partly inherited from KEEL, which distributes its datasets already divided into five folds: 13 of the 33 studies that use KEEL data use 5-fold cross-validation, against 7 of the 38 that do not. Five folds also leave more minority examples in each test fold: a dataset with 20 minority examples leaves 4 per fold with five folds and 2 with ten.

Resampling must be fitted on the training part of each split only; otherwise synthetic points are generated from test examples and the test set is no longer independent (Figure 5). Only 34 of the 71 studies (48%) state or describe unambiguously that resampling was confined to the training data, and 36 do not say. In one study the reported numbers of synthetic samples equal the full minority-class size, which implies that the whole dataset was oversampled before 5-fold cross-validation (Majzoub and Elgedawy, 2020); another appears to normalise features before splitting (Tao et al., 2025).

Stratification has a limit that matters for small minority classes. It keeps the class proportions equal across folds, but random assignment within a class can still put the examples of one region of the minority class in some folds and not in others, so that training and test folds differ in distribution; this is a form of dataset shift induced by the partition (Moreno-Torres et al., 2012). Distribution-optimally balanced stratified cross-validation (DOB-SCV) assigns neighbouring examples to different folds to prevent it, and on imbalanced data it gives a more stable estimate than standard stratified cross-validation (López et al., 2014). The fifteen-year review recommended it for SMOTE-based studies (Fernández et al., 2018); none of the 71 studies uses it.

Table 18: Validation Protocols

Protocol	Description	Advantages	Limitations	Studies	Examples
Repeated k-fold CV	k folds, reshuffled and repeated	Uses all data; variance can be estimated	Costly; folds overlap across repeats	33	(Napierala et al., 2010; Krawczyk et al., 2020; Naglik and Lango, 2024)
Single k-fold CV	One run of k folds	Uses all data once	One estimate per dataset; sensitive to the split	16	(Chawla et al., 2002; Fernandez et al., 2013; Maldonado et al., 2022)
5×2 CV	Five repeats of two folds	Suits paired tests on each dataset	Half the data for training	4	(Cieslak and Chawla, 2008; Koziarski et al., 2019)
Repeated hold-out	Several random train/test splits	Simple; variance can be estimated	Test sets overlap; few minority test examples	6	(He et al., 2008; Chen et al., 2021b)
Single hold-out	One train/test split	Cheap for large data	One estimate; high variance	8	(Vuttipittayamongkol and Elyan, 2020; Hu et al., 2026)
Fixed benchmark split	The split shipped with the dataset	Comparable with other work on that benchmark	One split; imbalance often induced artificially	3	(Buda et al., 2018; Mullick et al., 2019)

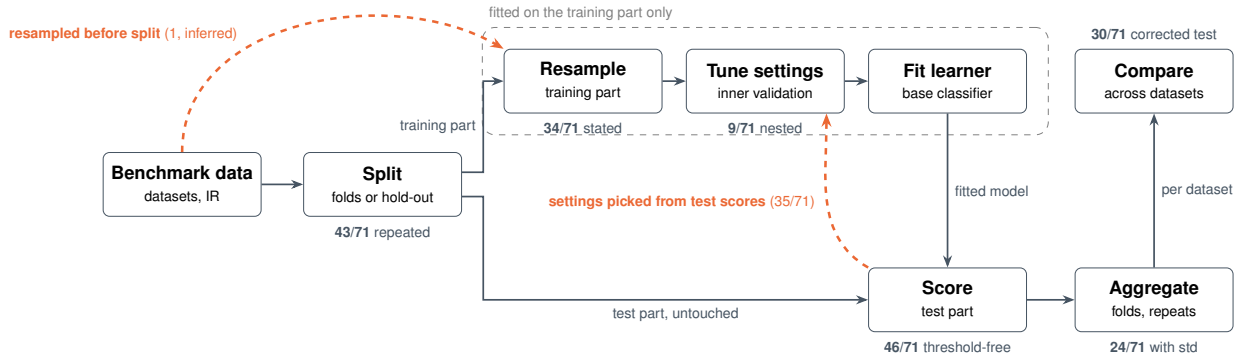


Figure 5: Where Resampling, Tuning and Testing Belong in One Run of an Experiment

Choosing settings. Every resampler has settings, such as the number of neighbours k or the amount of oversampling, and many learners do too. Choosing them is the weakest step in this literature (Figure 6). Only 9 studies (13%) tune inside the training data, by nested cross-validation or on a disjoint set of tuning datasets (Cieslak and Chawla, 2008; Nekooeimehr, 2016; Krawczyk et al., 2020; Yadav and Vijay, 2025). Twenty (28%) use fixed or default settings for every method, which is even-handed but can penalise methods with poor defaults. In 35 studies (49%) the reported configuration of the proposed method, or the amount of resampling, was chosen by looking at results on the evaluation data, typically by reporting the best point of a grid run on the same folds; 25 of these say so explicitly and 10 imply it by describing preliminary runs with no separate validation data (Barua et al., 2014; Abdi and Hashemi, 2016; Chen et al., 2021b). The remaining 7 do not say how settings were chosen. This practice has not declined: its share rose from 36% of the studies before 2010 to 54% since 2021, while the share that uses fixed settings for all methods fell from 57% to 21%.

What is chosen, and on what data? Forty-one studies choose parameters of their method, 20

Table 19: Protocol and Choice of Settings by Period

Practice	Studies (71)	–2009 (14)	–2015 (13)	–2020 (20)	–2026 (24)
Repeated k-fold CV	33	29%	46%	45%	58%
5×2 CV	4	7%	8%	5%	4%
Repeated hold-out	6	7%	15%	5%	8%
Single k-fold CV	16	36%	31%	10%	21%
Single hold-out	8	21%	0%	15%	8%
Fixed benchmark split or not stated	4	0%	0%	20%	0%
Repeated partitions (first three rows)	43	43%	69%	55%	71%
$k = 5$	20	0%	15%	25%	54%
$k = 10$	23	57%	62%	25%	8%
Train-only resampling stated	34	57%	38%	60%	38%
Settings chosen on evaluation data	35	36%	46%	55%	54%
Fixed or default settings for all	20	57%	31%	15%	21%
Nested or held-out tuning	9	7%	0%	20%	17%
Not stated	7	0%	23%	10%	8%

Period columns give the share of the studies of that period.

hyperparameters of the classifier, 14 the best of several variants of their method and 9 the amount of resampling; 24 choose nothing. The data used for the choice are the evaluation data in 25 studies, an inner validation split of the training data in 11 and a set of tuning datasets that are excluded from the evaluation in one (Yadav and Vijay, 2025); 19 do not say. Of the 54 studies that select a parameter value, a variant or an amount of resampling, 33 name the measure that decides: G-mean in 15, ROC-AUC in 12, F1 in 8 and AvAcc in 4 (Table 8). In all 33 the criterion is also one of the reported measures, so when the selection uses the evaluation folds, the proposal is tuned on exactly the number the reader sees. Recall alone decides only inside two internal rules (Das et al., 2015; Vuttipittayamongkol et al., 2018), and recall alone rewards a method that predicts everything as minority.

Choosing a model on the data that are then used to estimate its performance biases the estimate upwards (Cawley and Talbot, 2010), and the bias grows with the number of configurations tried, so it favours whichever method was searched most, which is usually the proposed one. Search budgets are often unequal: G-SMOTE is selected from 294 configurations against 2 for SMOTE (Douzas and Bacao, 2019). To see what this can do, suppose that every configuration is in truth equally good and that measured scores vary by a standard deviation of 0.02 around the true value. The best of 294 configurations then scores about 0.057 above the truth and the best of 2 about 0.011, so the method that was searched more looks better without being better. A principled alternative has existed since 2008, when a wrapper was proposed that chooses the amount of undersampling and SMOTE on the training data alone (Chawla et al., 2008).

What this means for a new study. Use stratified 5-fold cross-validation repeated at least five times with fixed seeds, the same partitions for every method and learner, or DOB-SCV when the minority class has few examples, and say that resampling, scaling, encoding, imputation and feature selection were fitted on the training folds only. Then either use the settings recommended by the authors of each method for every method, which is the safer choice for a comparison, or tune every method, including SMOTE and the other baselines, by nested cross-validation with the same search budget and a selection measure declared in advance. Never report the best point of a grid evaluated on the test folds; show the effect of a parameter as a separate sensitivity analysis instead.

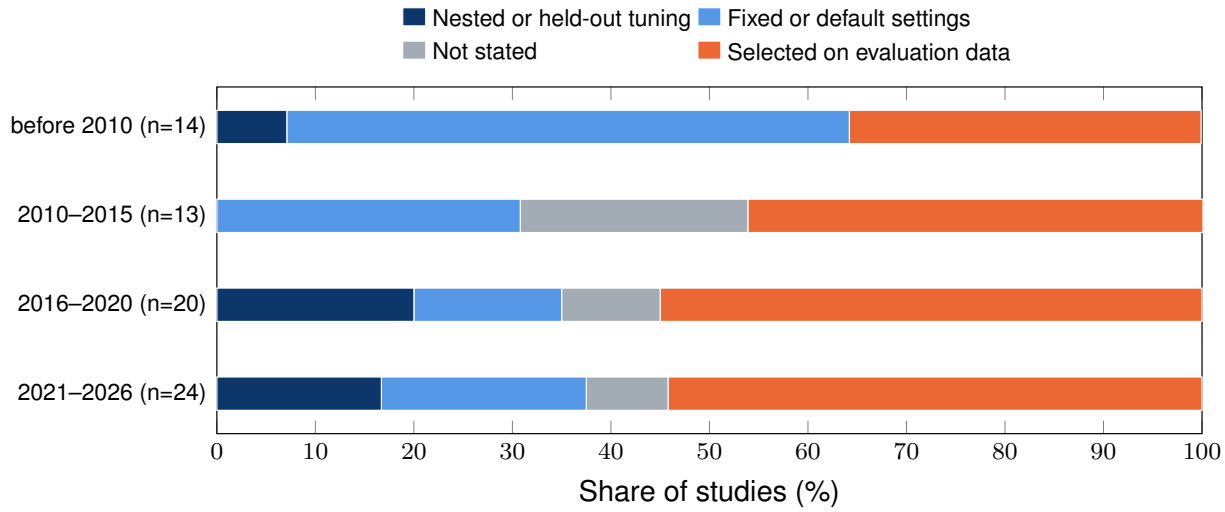


Figure 6: How Hyperparameters Are Chosen

Table 20: Statistical Tests by Period

Test	Studies (71)	–2009 (14)	–2015 (13)	–2020 (20)	–2026 (24)
Wilcoxon signed-rank	26	0%	46%	40%	50%
Friedman	25	7%	31%	45%	46%
Holm	10	7%	0%	15%	25%
Bonferroni–Dunn	6	0%	8%	10%	12%
Nemenyi	4	0%	15%	5%	4%
Shaffer	4	0%	0%	15%	4%
Iman–Davenport	4	0%	8%	5%	8%
t-test	7	21%	31%	0%	0%
Bayesian test	3	0%	0%	0%	12%
Cites Demšar (2006)	19	7%	46%	20%	33%
Cites any methodological source	25	7%	46%	30%	50%
No test	19	64%	8%	25%	17%

Period columns give the share of the studies of that period. A study can use several tests.

7.9 Statistical Testing

A significance test asks whether a difference across datasets is larger than chance would produce. Fifty-two studies (73%) use one, and the share without any test fell from 64% before 2010 to 17% since 2021 (Figure 7). The Wilcoxon signed-rank test (26 studies) and the Friedman test (25) dominate, with Holm (10), Bonferroni–Dunn (6), Shaffer (4) and Nemenyi (4) as post-hoc procedures (Table 20). The *t*-test, used by 21% of the studies before 2010 and 31% of those of 2010 to 2015, has not been used since. Grouped by what they control, 27 studies apply a corrected multiple-comparison procedure, in 23 cases a Friedman test followed by a post-hoc test against the proposed method; 19 use only pairwise tests without correction, although they compare the proposal with a median of six competitors; 3 stop at an omnibus test; 3 use Bayesian signed-rank tests (Zheng et al., 2022; Yan et al., 2025a,b); and 19 report no test.

Twenty-five studies cite a methodological source for their tests. The most cited is Demšar's guide to comparing classifiers over multiple datasets (Demšar, 2006) (19 studies), followed by Friedman's

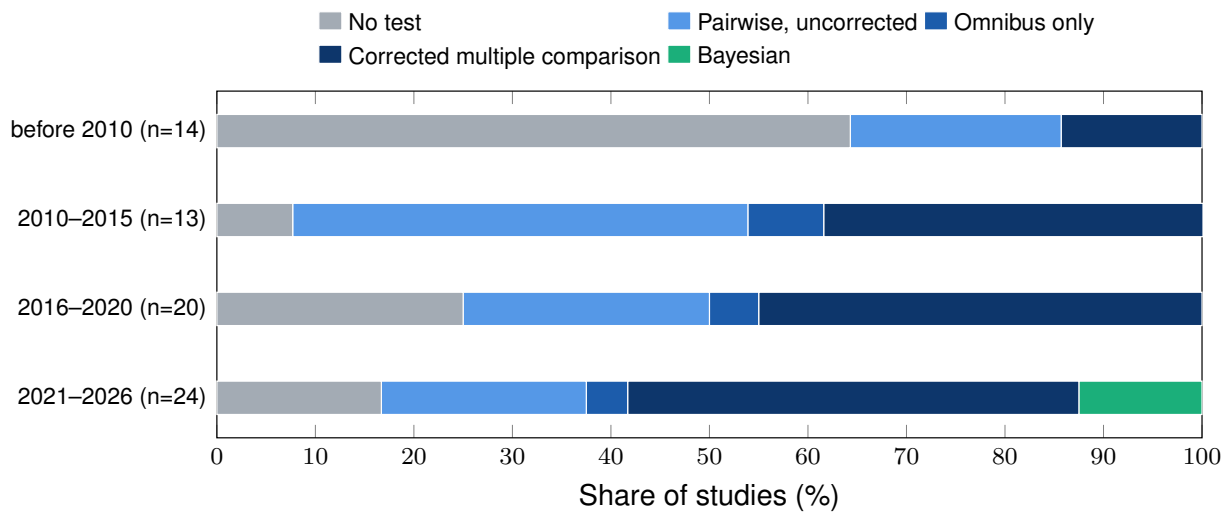


Figure 7: Statistical Testing by Period

Table 21: Errors Found in the Use of Statistical Tests

Study	Error
SOUP (2019) (Janicka et al., 2019)	A Nemenyi critical-difference diagram is drawn after a Friedman test that is not significant ($p = 0.36$)
MC-MBRC (2024) (Ma et al., 2024)	A Nemenyi test is applied without an omnibus Friedman test
MC-NCLWO (2025) (Tao et al., 2025)	The post-hoc procedure is named Holm in one place and Shaffer in another, and no omnibus test is reported
G-SMOTE (2019) (Douzas and Bacao, 2019)	Holm-adjusted p-values are reported, but the paired test behind them is not named
CBHSID (2022) (Palli et al., 2022)	Pairwise Wilcoxon tests without correction; the test is cited to a guide on normality testing; the significance level is given as 0.1 in one place and 0.01 or 0.05 in another

original papers (6 and 2), the tutorial of García and colleagues (García et al., 2010) (5), and Benavoli and colleagues for Bayesian tests (Benavoli et al., 2017) (3). The uncorrected pairwise tests matter because the chance of a false positive grows with the number of comparisons: ten comparisons at the 0.05 level without correction give at least one false positive with probability $1 - 0.95^{10} \approx 0.40$ when no method is truly better. Errors remain even among the studies that test (Table 21), such as a Nemenyi diagram drawn after a non-significant Friedman test (Janicka et al., 2019) and a Nemenyi test without an omnibus test (Ma et al., 2024).

Two limits of testing need stating. First, a significant test does not remove the selection bias of Section 7.8, because the tests are run on the same folds on which settings were chosen. Second, the unit of a test must be independent. The results of several learners on the same dataset are not independent, so ranks should be computed per learner over the datasets, or over the learner-averaged score of each dataset, rather than over pooled dataset-learner pairs, which treats five results on one dataset as five datasets and makes differences look more reliable than they are.

What this means for a new study. Make the dataset the unit of every test and test each learner and measure separately. For several methods, use the Friedman test with the Iman–Davenport

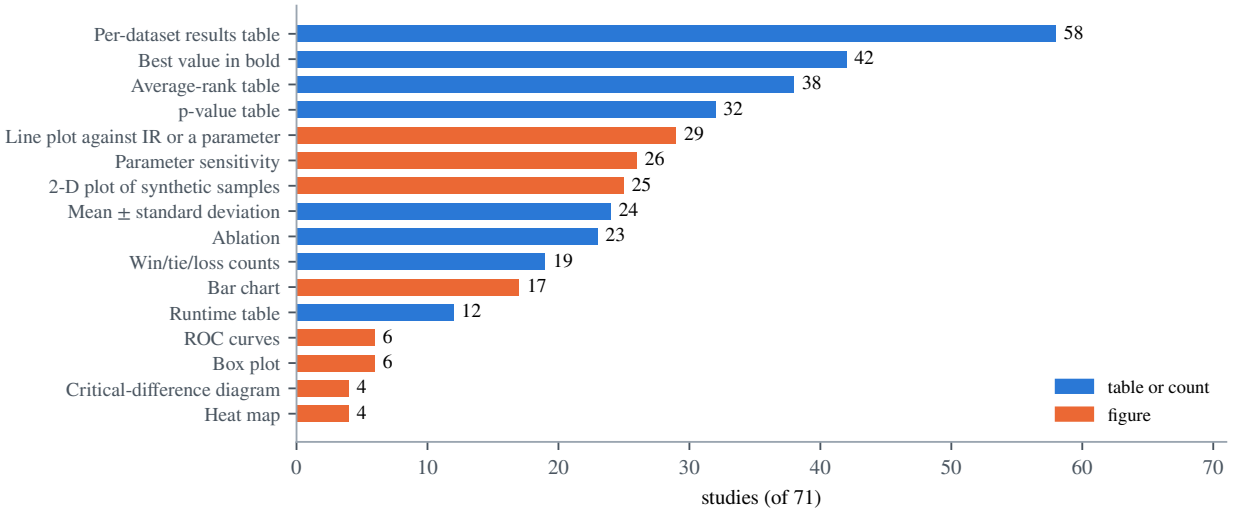


Figure 8: How the 71 Studies Report Their Results

correction followed by Holm's procedure against the proposal (or Shaffer's for all pairs), which have more power than the Nemenyi and Bonferroni–Dunn procedures (Demšar, 2006; García and Herrera, 2008); for two methods, the Wilcoxon signed-rank test. Report effect sizes, such as the median difference and win/tie/loss counts, next to the p-values, and to claim that two methods are equivalent in practice, use a Bayesian signed-rank test with a region of practical equivalence (Benavoli et al., 2017).

7.10 Reporting Results

How results are shown decides what a reader can check. Figure 8 counts what the studies show and Table 22 how this changed. Tables dominate: 58 studies give results per dataset, 42 mark the best value in bold, 38 give average ranks and 32 a table of p-values. Average ranks and p-values grew with significance testing (from 7% and 0% of the studies before 2010 to 75% and 67% since 2021), and so did ablation studies (14% to 50%), parameter-sensitivity analyses (7% to 71%) and tables in a supplement (0% to 42%). Standard deviations did not: they are given by 24 studies, between 29% and 38% in every period.

Four gaps stand out. Thirty-eight studies compute average ranks, but only 4 draw the critical-difference diagram that shows at a glance which methods cannot be told apart. Forty-two mark the best value in bold, but only 24 give a standard deviation, so in most tables a reader cannot tell whether a bold value differs from the next one by more than the variation between runs. Seventeen show bar charts, which, when used for averages across datasets, mix values on different scales so that easy datasets dominate. And 6 show ROC curves, which can be read for a few datasets but not for 30 to 100.

Table 23 pairs each purpose of a results section with what the corpus does and what we recommend. A results table should give the mean and standard deviation over repeated runs, an imbalance-ratio column with the datasets sorted by it, and summary rows with the average rank, win/tie/loss counts against the main baseline and Holm-adjusted p-values; the full per-dataset table belongs in a supplement, and the main text needs only the summary, together with a statement of what bold means and how ties are handled. Table 24 shows this form for one learner and one measure of our illustrative runs. The figures should be a critical-difference diagram for every mea-

Table 22: Forms of Reporting by Period

Form	Studies (71)	–2009 (14)	–2015 (13)	–2020 (20)	–2026 (24)
Results table per dataset	58	71%	100%	80%	79%
Best value in bold	42	43%	69%	50%	71%
Average-rank table	38	7%	46%	65%	75%
Table of p-values	32	0%	46%	50%	67%
Mean ± standard deviation	24	29%	38%	30%	38%
Win/tie/loss counts	19	36%	15%	25%	29%
Ablation	23	14%	23%	30%	50%
Parameter sensitivity	26	7%	15%	30%	71%
Critical-difference diagram	4	0%	15%	5%	4%
Box plot	6	7%	8%	0%	17%
Runtime table	12	7%	15%	15%	25%
Tables in a supplement	18	0%	15%	30%	42%
Running time reported (table or figure)	14	14%	15%	15%	29%
Code released	27	0%	15%	65%	50%

Period columns give the share of the studies of that period.

sure reported, per learner or as separate panels; a heat map of average ranks when there are several measures and learners; a per-dataset plot of each method against the main baseline, which shows both the size of the gain and how often the method loses; the gain against the imbalance ratio or the size of the data, which shows where the method helps; and running time against dataset size on log axes.

What this means for a new study. For each measure, give one summary table with average ranks per learner, win/tie/loss counts against the main baseline and adjusted p-values, and one critical-difference diagram; put per-dataset means and standard deviations in a supplement; add a heat map of ranks when there are several measures and learners, the per-dataset gain against the imbalance ratio, and running time. Release the code, seeds, fold indices and per-fold results with the library versions, so that others can re-analyse the results with another test or measure and can use the method as a competitor. Avoid bar charts of averages across datasets and ROC curves for more than a few datasets.

7.11 Why Evaluation Practice Changed

Section 7.2 explained the change of each learner and measure. Here we step back and ask what kind of event makes evaluation practice change at all. Table 25 lists the drivers with the evidence for each. The evidence that links a driver to a change is of three kinds: the first use of the practice comes after the driver appeared, the studies that adopt the practice cite the driver, and the studies that use the driver behave differently from those that do not, as when the users of KEEL data choose five folds more often.

Practices spread when something made them cheap and expected at the same time. Significance testing is the clearest case. Demšar's guide to comparing classifiers over multiple datasets (Demšar, 2006) gave a complete recipe, KEEL and later tutorials implemented the Friedman test with Holm and Shaffer post-hoc procedures (García and Herrera, 2008; García et al., 2010; Derrac et al., 2011), and reviewers began to expect it. The first Friedman test in the corpus appears in 2008 and the first Wilcoxon test in 2010; the last t-test appears in 2015; the share of studies without any test fell from 64% before 2010 to 17% since 2021; and 19 studies cite Demšar for their tests. Multi-dataset

Table 23: How Results Are Reported and What We Recommend

Purpose	In the corpus (of 71)	Recommended	Why
Results per dataset	58 studies give a table, 24 with a standard deviation	Mean ± standard deviation over repeated runs, datasets sorted by IR, full table in the supplement	The reader can see whether a gain exceeds the run-to-run variation
Summary over datasets	38 give average ranks, 19 win/tie/loss counts	Average ranks per learner, win/tie/loss against the main baseline, adjusted p-values, in the same table	One table answers "how often, by how much and is it reliable"
Significance	4 draw a critical-difference diagram	One critical-difference diagram per measure (and per learner, or per learner-averaged score)	Shows at a glance which methods cannot be told apart
Several measures and learners	4 heat maps	A heat map of average ranks, measures by learners	Shows whether the conclusion depends on the measure
Where the method helps	29 plot a measure against IR or a parameter	Per-dataset gain over the baseline against IR or dataset size	Tells a user on which data the method is worth its cost
Cost	12 runtime tables	Running time against dataset size on log axes	Separates a practical method from a slow one
Figures to avoid	17 bar charts, 6 ROC plots	Bar charts of means across datasets; ROC curves for dozens of datasets	Averages mix scales; overlaid curves cannot be read

benchmarks spread in the same way after the KEEL repository made dozens of pre-partitioned imbalanced datasets available (Alcalá-Fdez et al., 2009, 2011): KEEL data first appear in 2013 and are used by 67% of the studies since 2021, and studies with fewer than ten real datasets fell from 64% to 4%. Code release and the modern learners of Section 7.2 followed scikit-learn and imbalanced-learn (Lemaître et al., 2017). Other changes followed a change of problem rather than a tool: multi-class measures spread when the family moved to multi-class problems, and image data and convolutional networks came and went with a few deep-learning studies. Some changes are carried by lines of work, as with the set of multi-class measures that every study of the MC-CCR line reports. And one change followed from the mechanism of Section 7.6.1: G-mean is the measure under which a resampler shows its effect most clearly. In short, practice changed when there was a recipe that could be followed at once, shared data and tools that made it cheap, a new problem that required it, or a measure that showed the effect of the proposal.

Being well known is not enough. PR-AUC is recommended for imbalanced data by a paper with more than 5,000 citations (Saito and Rehmsmeier, 2015) and is still used by fewer than one study in ten. Bayesian tests, recommended in 2017 (Benavoli et al., 2017), appear in three studies from 2022 on. The equivalence of preprocessing and cost-sensitive learning (López et al., 2012) and DOB-SCV (López et al., 2014) were both restated in the fifteen-year review, itself cited more than 2,000 times (Fernández et al., 2018), and neither changed the comparisons that followed: cost-sensitive competitors remain rare and DOB-SCV is never used. And the weakest step, choosing settings on evaluation data, became more common rather than less, although the bias it causes has been described since at least 2010 (Cawley and Talbot, 2010). These practices have no tool that makes them the default and no benchmark that requires them.

Citations. Nor are careful studies visibly rewarded with citations. Table 26 gives the median citation count of the studies with and without each practice, with the median year of each group; as Section 7.1 explained, a difference in counts has to be read together with the difference in years. Read this way, most differences are differences of age. Studies that use KEEL data, many datasets, a significance test, a corrected test or a methodological source have fewer citations than studies

Table 24: A Results Table in the Recommended Form (Illustrative Runs, SVM, G-mean)

Dataset	IR	None	SMOTE	MWMOTE	CISO	QC-SMOTE
wisconsin	1.9	0.966 ± 0.007	0.971 ± 0.008	**0.974 ± 0.006**	0.971 ± 0.003	0.970 ± 0.003
hepatitis	3.8	0.625 ± 0.149	0.712 ± 0.098	**0.719 ± 0.077**	0.715 ± 0.101	0.691 ± 0.095
yeast_0_2_5_6_vs_3_7_8_9	9.1	0.621 ± 0.064	0.783 ± 0.047	0.742 ± 0.026	**0.786 ± 0.046**	0.786 ± 0.050
ecoli_0_1_vs_5	11.0	0.839 ± 0.227	**0.904 ± 0.129**	0.879 ± 0.161	0.881 ± 0.158	0.839 ± 0.227
glass_0_1_6_vs_5	19.4	0.283 ± 0.387	0.816 ± 0.154	0.816 ± 0.154	**0.819 ± 0.153**	**0.819 ± 0.153**
kddcup_guess_passwd_vs_satan	30.0	0.950 ± 0.050	0.989 ± 0.023	0.949 ± 0.050	0.949 ± 0.050	**0.999 ± 0.001**
abalone_19_vs_10_11_12_13	49.7	0.000 ± 0.000	**0.601 ± 0.103**	0.436 ± 0.285	0.563 ± 0.149	0.365 ± 0.220
poker_8_vs_6	85.9	0.720 ± 0.198	0.941 ± 0.131	**1.000 ± 0.000**	0.915 ± 0.130	0.915 ± 0.130
102 further datasets in the supplement						
Average rank ($N = 110$)		4.63	**2.10**	2.56	3.01	2.71
Win/tie/loss against SMOTE		5/6/99	control	33/19/58	21/16/73	29/24/57
Holm-adjusted p (Wilcoxon)		<0.001	–	0.003	<0.001	<0.001

Mean ± standard deviation over the five folds of a single stratified 5-fold run; bold marks the best value in a row. Friedman test (Iman–Davenport): $p = 1.5 \times 10^{-43}$. Datasets were chosen to span the range of IR. CISO and QC-SMOTE are our reimplementations.

without them, but they are also younger, by two to seven years. Where the two groups are of similar age, the differences are small and point both ways: studies that choose their settings on the evaluation data have a median of 129 citations against 152 (median years 2019 and 2018), and within three of the four periods they are cited more, not less; studies that report a standard deviation have 129 against 148 at the same median year. Releasing code goes with fewer citations overall (122 against 208), but the code releasers are five years younger, and within the periods the comparison points both ways. Only three practices point the same way in every period. Including SMOTE as a competitor goes with more citations (for example 238.5 against 75.5 among the studies of 2016 to 2020), a measure-definition problem with fewer (Section 7.6), and so, unexpectedly, does a no-resampling baseline, although in the first two periods the group without it has only two studies, one of which is SMOTE itself. The ten most cited studies show the same picture from the other side (Table 27): they use a median of 8 real datasets, only 4 of them use any significance test, and 7 were published before 2010. Citations follow age and the problem a paper addressed, not the care of its evaluation. In short, the field has improved where tools and reviewers pushed it, and not where improvement would have required authors to give up an advantage.

7.12 What the Comparisons Conclude

An outcome is what a study concludes about its own proposal (Table 28). Of the 69 studies that propose a method, 68 report it as the best overall: 27 with no stated exception and 41 with an exception they state themselves, such as a measure, learner, competitor or data condition under which the proposal is not better or not significantly better. Only one reports no advantage over simple baselines (Yadav and Vijay, 2025), and that study is also the only one that registered its protocol in advance. When almost every study reports that its method is the best, the claim carries little information; a statement of when the method is better is more useful.

These wins do not add up to a ranking. A method that leads in one study is placed below the proposal in every later study that includes it: MDO is a competitor in 13 later studies and MC-CCR in 7, and each of those studies reports its own method ahead on its main summary measure (Naglik and Lango, 2024; Yan et al., 2025b; Ma et al., 2024). Because every paper chooses its own measures, folds and settings, and usually tunes only its own method on the evaluation data, the corpus cannot say which of its methods is better; it can only say that each is better under the protocol of the paper that proposed it.

Table 25: What Drove the Changes in Evaluation Practice

Driver	Evidence by period (1997–2009 → 2010–2015 → 2016–2020 → 2021–2026)	Changes it explains
Change of toolkit	Studies naming Python: 0 → 0 → 6 → 12; naming Weka: 1 → 6 → 4 → 1; releasing code: 0 → 2 → 13 → 12	C4.5 falls and CART rises; rule learners disappear; logistic regression, random forest and gradient boosting arrive; MCC and macro-F1 appear
Absorption into composites	Composite users who also report recall: 8/9 → 5/21; F users who report precision: 5/7 → 2/10; G-mean users who report TNR: 2/4 → 0/17	Recall, precision and TNR fall or vanish as stand-alone measures; G-mean rises
Larger benchmarks	Median datasets per study: 6.5 → 20 → 20 → 23	ROC curves and confusion matrices disappear; running time is reported more often
Change of problem or data	Multi-class studies: 3 → 4 → 10 → 15; image studies: 0 → 0 → 2 → 0; generative studies from 2019 (5)	MAUC, MG and macro-F1 rise; CNNs come and go; accuracy survives in generative studies
Following a line of work	11 studies say they chose learners as an earlier study did; six studies of the MC-CCR line (2020 to 2025) share AvAcc, CBA and CEN	CBA, CEN, AvAcc and MLP spread together; C4.5 persists where comparability with older work matters
Methodological sources	Demšar (2006) cited for the tests by 19 studies; first Friedman test 2008, first Wilcoxon test 2010; last t-test 2015	Significance testing spreads (studies without a test: 64% → 17%)
Realism and critiques	RF or GB: 7% → 33%; critiques of F1 and accuracy (Boughorbel et al. cited by 3 studies)	Strong learners appear; MCC appears; F1 stops growing

A minimum standard. To summarise how far the evaluations are from the recommendations of this section, we scored every study against seven points that are cheap to meet and that a reviewer can check from the paper: repeated partitions, a statement that resampling is confined to the training data, settings that are not chosen on the evaluation data, a corrected or Bayesian test, a no-resampling baseline, at least one composite and one ranking measure, and released code (Table 29). The points met least often are released code (27 studies), settings not chosen on the evaluation data (29), a corrected or Bayesian test (30) and a composite together with a ranking measure (32). The median study meets 3 of the seven points, and the median rose only from 3 before 2010 to 4 since 2021. One study meets all seven, RBO (Koziarski et al., 2019); four meet six (Douzas et al., 2018; Douzas and Bacao, 2019; Ren et al., 2025, 2026), and 39 meet three or fewer. The score is a summary of practice, not a judgement of the quality of any single study.

7.13 A Protocol for a New SMOTE-Family Study

The following protocol collects the recommendations of this section, topic by topic. Each step says what to choose and why; Table 30 lists what the paper should then report, so that a reader can reproduce the comparison and combine it with others.

1. **State the question and the primary measure before running anything.** Say what kind of problem the method is for (binary or multi-class, tabular, the range of imbalance) and which measure decides success, chosen from the costs of the application (Section 7.6.2). Why: the choice of measure alone can decide the winner.

2. **Choose data that test the claim and can be shared.** Use the imbalanced datasets of KEEL as the core, at least 30 of them, covering imbalance ratios from about 1.5 to more than 100 and different sizes and dimensions; add four to eight controlled synthetic datasets

Table 26: Citations of Studies With and Without Each Practice

Practice	With the practice			Without		
	Studies	Citations	Year	Studies	Citations	Year
Uses KEEL data	33	100	2020	37	233	2013
At least 18 real datasets (the median)	36	94.5	2021	34	267.5	2014
States why the data were chosen	58	140.5	2019	12	320.5	2015.5
More than one base learner	46	157	2019	24	136	2018
Random forest or boosting learner	22	135.5	2020	48	145.5	2017
States why the learners were chosen	48	111	2019.5	22	291	2013
Includes a no-resampling baseline	51	152	2018	19	122	2022
Includes SMOTE as a competitor	50	150	2019.5	20	84.5	2018
Reports accuracy or error rate	13	229	2014	57	129	2019
Reports a threshold-free measure	45	148	2018	25	129	2019
Measure-definition problem	14	25	2022	56	218.5	2017.5
Measures resampled data directly	6	238.5	2019.5	64	140.5	2018
Repeated partitions	42	133.5	2019	28	205	2018.5
States train-only resampling	34	164.5	2018	36	130	2020
Settings chosen on evaluation data	35	129	2019	35	152	2018
Any significance test	51	122	2019	19	303	2013
Corrected or Bayesian procedure	30	111	2020	40	218.5	2016.5
Cites a testing methodology	25	100	2020	45	152	2018
Reports standard deviations	23	129	2019	47	148	2019
Reports running time	14	99.5	2021	56	159	2018
Code released	27	122	2020	43	208	2015
States where proposal is not better	41	177	2016	29	93	2020

Citations: median citation count in OpenAlex on 18 October 2026 (70 of the 71 studies; one has no record). Year: median year of publication of the group. Older papers have had more time to be cited (the median is 1,738 citations for studies published up to 2009 and 21 for studies since 2021), so a difference between the two groups must be read together with the difference in their years.

that vary overlap, imbalance and small disjuncts one factor at a time; measure the difficulty of each dataset, for example by the shares of safe, borderline, rare and outlying minority examples (Napierała and Stefanowski, 2016), and use it to explain where the method wins; keep multi-class data multi-class; and release the processed files and folds. Why: the dataset is the unit of the statistical test, so more datasets give more reliable conclusions (Demšar, 2006); the imbalance ratio alone does not describe difficulty, which depends more on overlap, small disjuncts and noise (López et al., 2013); and 42 of the 52 studies that modify their data do not release the result (Section 7.3).

3. **Use several learners of different kinds.** At least four: a decision tree (CART or C4.5), k-NN, a linear or margin-based model (logistic regression or an SVM) and a strong ensemble (random forest or gradient boosting), configured identically for every resampler, and the learners of the papers compared with. Why: the effect of resampling depends on the learner, strong learners reduce it, and with a single learner a reader cannot tell a better method from a good fit to that learner (Section 7.4).

4. **Include the baselines a reviewer will ask for.** No resampling, class weighting or threshold moving of the learner, random oversampling and plain SMOTE, plus the two to four strongest recent competitors, all re-run on the same folds with maintained or verified implementations. Why: threshold measures reward any shift of the operating point (Section 7.6.1), cost-sensitive learning has been found as effective as preprocessing (López et al., 2012), and copied numbers come from other folds and settings (Section 7.5).

Table 27: How the Ten Most Cited Empirical Studies Evaluate

Study	Citations	Real datasets	Learners	Protocol	Test
SMOTE (2002) (Chawla et al., 2002)	32,518	9	3	single k-fold	none
ADASYN (2008) (He et al., 2008)	4,927	5	1	repeated hold-out	none
Borderline-SMOTE (2005) (Han et al., 2005)	4,313	3	1	repeated k-fold	none
Batista et al. (2004) (Batista et al., 2004)	4,207	15	1	single k-fold	corrected
Buda et al. (2018) (Buda et al., 2018)	3,053	3	1	fixed split	none
OSS (1997) (Kubat and Matwin, 1997)	2,562	7	2	single k-fold	none
EasyEnsemble (2009) (Liu et al., 2009)	2,475	16	2	repeated k-fold	pairwise
RUSBoost (2010) (Seiffert et al., 2010)	1,867	15	3	repeated k-fold	corrected
SMOTEBoost (2003) (Chawla et al., 2003)	1,738	4	1	single k-fold	none
k-means SMOTE (2018) (Douzas et al., 2018)	1,273	80	3	repeated k-fold	corrected

Table 28: Outcomes Reported by the Studies for Their Own Proposals

Outcome	Description	Studies	Examples
A. Best, no stated exception	The proposal is reported best on all summary measures and learners	27	(Douzas et al., 2018; Chen et al., 2021a; Yan et al., 2025b)
B. Best, with stated exceptions	Best overall, but the authors name a measure, learner, competitor or data condition where it is not better	41	(Seiffert et al., 2010; Janicka et al., 2019; Zhu et al., 2023)
C. No advantage over simple baselines	Not better than no resampling, class weighting or random oversampling	1	(Yadav and Vijay, 2025)
D. Comparative study	No new method; compares existing ones	2	(Buda et al., 2018; Fernandez et al., 2013)

5. **Report measures of both kinds, and define them.** For two classes, ROC-AUC and PR-AUC computed from scores, together with G-mean (or balanced accuracy) and F1 or MCC at the default threshold; for several classes, MAUC with macro-F1 or AvAcc and MG. Give the formula, the positive class, the threshold and the averaging rule of each, and do not use accuracy or weighted F1 as the primary measure. Why: G-mean and balanced accuracy reward predicting the minority class, F1 and MCC penalise false alarms, and only the two kinds together show whether a gain comes from a moved threshold or from a better ranking (Section 7.6).

6. **Keep the test data untouched.** Stratified 5-fold cross-validation repeated at least five times with fixed seeds (or DOB-SCV when minority classes are small), the same partitions for every method, and every learned step, from resampling to feature selection, fitted on the training folds only. Why: one run gives one noisy estimate per dataset, and resampling before splitting puts information from the test data into training (Section 7.8).

7. **Choose settings fairly.** Use the settings recommended by the authors of every method, or tune every method with the same budget by nested cross-validation and a selection measure declared in advance; never report the best point of a grid on the test folds. Why: selection on the evaluation data inflates the score of the method searched most (Section 7.8).

8. **Test the differences correctly.** Datasets as the unit, each learner and measure sepa-

Table 29: A Seven-Point Minimum Standard and How Many Studies Meet It

Point	Studies (71)	–2009 (14)	–2015 (13)	–2020 (20)	–2026 (24)
Repeated partitions	43	43%	69%	55%	71%
Train-only resampling stated	34	57%	38%	60%	38%
Settings not chosen on evaluation data (nested or fixed)	29	64%	31%	35%	38%
Corrected or Bayesian test	30	14%	38%	45%	58%
No-resampling baseline	52	86%	85%	80%	54%
A composite and a ranking measure	32	14%	46%	45%	62%
Code released	27	0%	15%	65%	50%
Median number of points met	3	3	3	3.5	4
All seven points met	1	0	0	1	0

Period columns give the share of the studies of that period that meet the point, except in the last two rows.

rately; Friedman with Iman–Davenport and Holm against the proposal (Shaffer for all pairs), Wilcoxon for two methods, effect sizes next to p-values, and Bayesian signed-rank tests with a region of practical equivalence to claim that two methods are equal. Why: uncorrected pairwise tests and pooled dataset-learner pairs make chance differences look significant (Section 7.9).

9. **Check the resampled data directly.** At least one check on held-out data, such as the share of synthetic points whose neighbours are mostly majority examples, a two-sample test against held-out minority examples, or the shares of safe, borderline, rare and outlying examples before and after resampling, related to the difficulty of the data. Why: scores show that a classifier improved, not why, and not whether the synthetic points fall in the wrong place (Section 7.7).

10. **Report so that results can be read and reused.** Summary tables with ranks, win/tie/loss counts and adjusted p-values; critical-difference diagrams; per-dataset means and standard deviations in a supplement; gain against imbalance ratio; running time; code, seeds, folds and per-fold results (Section 7.10). Why: without standard deviations a reader cannot separate a real difference from the variation between folds, and without code a method is rarely used as a competitor.

11. **State the scope, the exceptions and the limits.** Give conclusions per measure and learner, including where the method loses, with effect sizes and practical equivalence rather than significance alone; state the limits of the study (own implementations, chosen settings, a single source of data, what was not tested); say that the conclusions hold for classification (Section 7.3.1); and register the plan in advance where possible (Yadav and Vijay, 2025). Why: when almost every study reports that its method is the best, only conditional conclusions carry information (Section 7.12).

Our own illustrative runs (Appendix A) show what this protocol asks of a real experiment. They already meet ten of its requirements: 110 KEEL datasets with their imbalance ratios, no-resampling and SMOTE baselines, five learners from different families reported separately, ROC-AUC and PR-AUC from scores together with F1 and G-mean, stratified 5-fold cross-validation on the same folds with fixed seeds, resampling and scaling fitted on the training folds only, the same fixed settings for every method, per-fold results kept, average ranks with Friedman tests and win counts,

Table 30: What a New Study Should Report

Topic	What to report
Benchmark data	Source and name of every dataset (KEEL or UCI) with the date of download; size, number of features, number of classes and IR of each, with the definition of IR; how classes were merged, categorical features encoded and missing values handled; why these datasets (range of IR and difficulty, not only popularity); where the processed files and fold assignments are
Base learners	Name, library and version of each learner; every hyperparameter and whether it was tuned; which score feeds AUC (probability or decision function); why each learner was chosen
Competitors and baselines	Every competitor with the source of its code (library and version, or own implementation); the parameters of each; a statement that all were re-run on the same folds; for own implementations, which part of the original paper was followed, how unspecified values were set and how the implementation was checked
Measures	The formula of every measure and the positive class; the threshold of each threshold measure; the scores used for AUC and PR-AUC; the averaging rule for several classes; the primary measure and why; the function or library that computes it
Validation protocol	k, the number of repetitions, stratification and seeds; which steps are fitted on the training folds only; a statement that every method uses the same partitions; the smallest number of minority examples in a test fold
Choice of settings	The settings or search grid of every method; the search budget of each (number of configurations tried); the measure and data used to choose; a statement that nothing was chosen on the test folds
Statistical testing	The test, its unit (the dataset) and N; the significance level and the correction for multiple comparisons; the measures and learners tested; the software; a table of p-values or a critical-difference diagram; effect sizes
Reporting	Which tables are in the main text and which in the supplement; where code, per-fold results and data files are; library versions and the hardware used for timing
Direct checks of the data	The definition of each check and the data it is computed on; how its results relate to the classifier scores; two-dimensional plots used as illustrations only
Claims and limits	Conclusions per measure and learner, including where the method loses; effect sizes, not only p-values; the limits of the study (own implementations, chosen settings, data sources, what was not tested); a link to a pre-registered plan if there is one

and conclusions stated per measure. They lack eleven: repeated cross-validation, class-weighting and random-oversampling baselines, a gradient-boosting learner, MCC or balanced accuracy and a declared primary measure, Friedman and Holm tests per learner over the 110 datasets throughout, effect sizes and Bayesian tests, standard deviations, running time and critical-difference diagrams in every report, direct checks of the synthetic points and of dataset difficulty, controlled synthetic data, a check of the re-implemented CISO and QC-SMOTE against the numbers of their papers, and released code, folds and results. That an experiment designed with these recommendations in mind still misses half of them is a measure of how much the protocol asks, and of why a checklist helps.

7.14 Limitations of This Analysis

Our analysis has six limits. First, the corpus is the SMOTE25 collection, chosen for its relevance to SMOTE rather than as a random sample of imbalanced-learning papers, so shares describe this corpus and not the field as a whole; it is the same list of papers, with the same 37 SMOTE-family members, that the Method and Data sections of this survey use [TODO: references to the Method and Data sections], and statements about individual datasets are left to the Data section. Second, the periods hold 13 to 24 studies each, so a single study moves a share by 4 to 8 points, and only large changes should be read as trends. Third, the causes we give are supported by direct or circumstantial evidence or are conjectures, as stated in each case; with observational data from 71 papers none of them is a causal estimate, and counts of named software are lower bounds because many studies do not name their tools. Fourth, citation counts depend on age, venue and topic as much as on the evaluation; we report them raw, with the years of the groups compared, and they show association only. Fifth, the coding was done by one reader with a second, page-referenced pass that a second reader has not yet checked. Sixth, the illustrative runs use a single cross-validation run, re-implementations of CISO and QC-SMOTE and no class-weighting baseline; they illustrate the form of a report and the effect of resampling on different measures, not the merit of any sampler.

A Illustrative Runs and the Coding of the Studies

Illustrative runs. Section 7 uses one small experiment in two places: to show how resampling affects different measures (Figure 4) and to show what a recommended results table looks like (Table 24). We resampled 110 binary imbalanced datasets of the KEEL repository (Alcalá-Fdez et al., 2011) with no resampling, SMOTE (Chawla et al., 2002), MWMOTE (Barua et al., 2014), CISO (Yadav and Vijay, 2025) and QC-SMOTE (Upman and Gowda, 2026), and trained five learners on the result: CART, k-NN ($k = 5$), an SVM with an RBF kernel, logistic regression and a random forest, with the same fixed scikit-learn settings for every sampler. Performance was estimated by one run of stratified 5-fold cross-validation with fixed seeds and the same folds for every sampler and learner; oversampling and feature scaling were fitted on the training folds only, and ROC-AUC and PR-AUC were computed from predicted probabilities or, for the SVM, from the decision function. Every sampler ran with the settings recommended by its authors; CISO and QC-SMOTE are our own re-implementations. These runs illustrate the form of a report and the effect of resampling on different measures; with a single cross-validation run and no class-weighting baseline they say nothing about the merit of any sampler [TODO: repeat the cross-validation at least five times if the runs stay in the paper].

Coding of the studies. Table 31 gives the main codes of every empirical study, so that each count in Section 7 can be traced to the studies behind it. The full coding, with page references and

verbatim quotes for the second reading, is in the supplementary material.

Table 31: Coding of the 71 Empirical Studies

Study	Set.	Data	Learners	Measures	Prot.	Settings	Test	Code	Std.
OSS (1997) (Kubat and Matwin, 1997)	B	7 U	kNN C45	GM TPR TNR Acc	CV*	Fixed	–		3
SHRINK (1998) (Kubat et al., 1998)	B	1	C45 kNN	GM TPR TNR	RCV*	Eval	Pair		3
SMOTE (2002) (Chawla et al., 2002)	B	9 U	C45 Rule NB	AUC TPR TNR	CV	Eval	–		0
SMOTEBoost (2003) (Chawla et al., 2003)	B+M	4 U	Rule	Prec TPR Fb	CV	Eval	–		1
Batista et al. (2004) (Batista et al., 2004)	B	15 U	C45	AUC	CV	Fixed	Corr		3
Borderline-SMOTE (2005) (Han et al., 2005)	B	3 U	C45	TPR F1	RCV*	Fixed	–		4
Cluster-SMOTE (2006) (Cieslak et al., 2006)	B	2	Rule	TPR FPR	HO*	Eval	–		1
Global-CS (2006) (Zhou and Liu, 2006)	M	14 U	C45	Cost MAUC	RCV	Fixed	Pair		3
COG-OS (2007) (Wu et al., 2007)	B+M	15 U	SVM LR C45 Rule	Prec TPR F1 Acc PCR	HO*	Eval	–		2
ADASYN (2008) (He et al., 2008)	B	5 U	CART	Acc Prec TPR F1 GM	RHO*	Fixed	–		4
LS (2008) (Cieslak and Chawla, 2008)	B	20 U	C45 SVM	AUC	5x2*	Nest	Corr		5
EasyEnsemble (2009) (Liu et al., 2009)	B	16 U	Boost RF	AUC F1 GM	RCV*	Fixed	Pair		5
SBC (2009) (Yen and Lee, 2009)	B	2 U	MLP	Prec TPR F1	HO	Fixed	–		2
Safe-Level-SMOTE (2009) (Bunkhumpornpat et al., 2009)	B	2 U	C45 NB SVM	Prec TPR F1 AUC	CV	Fixed	–		3
Global-CS (2010) (Zhou and Liu, 2010)	M	14 U	C45 MLP Boost	Cost MAUC	RCV	Fixed	Pair		3
RAMOBoost (2010) (Chen et al., 2010)	B	19 U	MLP	Acc Prec TPR Fb GM AUC	RHO	NS	Pair		2
RUSBoost (2010) (Seiffert et al., 2010)	B	15 U	C45 Rule NB	AUC PRAUC F1	RCV	Fixed	Corr		5
SPIDER2 (2010) (Napierala et al., 2010)	B	9 U	Rule C45	TPR TNR GM Acc	RCV	Fixed	Pair		3
DSRBF (2011) (Fernandez-Navarro et al., 2011)	B+M	13 U	MLP	Acc MS	RCV*	NS	Corr		4
INOS (2013) (Cao et al., 2013)	B	7	SVM	F1 GM Prec TPR TNR	RHO*	Eval	–		3
Fernandez et al. (2013) (Fernandez et al., 2013)	M	24 K	C45 SVM kNN	AvAcc Fb	CV*	Eval	Pair	yes	3
MWMOTE (2014) (Barua et al., 2014)	B	20 U	MLP Boost kNN C45	Acc Prec TPR F1 GM AUC	CV	Eval	Pair		1
RACOG (2015) (Das et al., 2015)	B	10 U	C45 SVM kNN LR	TPR GM AUC	CV*	Eval	Pair		3
Random Balance (2015) (Diez-Pastor et al., 2015)	B	86 KU	C45 kNN SVM	AUC F1 GM Acc Kap	5x2*	NS	Corr		5
SCUT (2015) (Agrawal et al., 2015)	M	8 KU	C45 SVM NB kNN	GM Fb AUC	CV	Fixed	Omni		3

Study	Set.	Data	Learners	Measures	Prot.	Settings	Test	Code	Std.
SMOTE-IPF (2015) (Saez et al., 2015)	B	9 K	C45	AUC	RCV	Eval	Corr	yes	4
SplitBal (2015) (Sun et al., 2015)	B	46 K	NB C45 Rule RF SVM kNN	AUC	RCV	Eval	Corr		3
A-SUWO (2016) (Nekooeimehr, 2016)	B	16 U	SVM LR kNN LDA	Fb GM AUC Prec TPR	RCV*	Nest	Corr		5
CGMOS (2016) (Zhang et al., 2016)	B	30 U	Oth kNN SVM MLP RF Boost	Prec TPR F1 GS AUC	RCV	NS	Pair		3
MDO (2016) (Abdi and Hashemi, 2016)	M	20 KU	C45 Rule kNN	MAUC GM Prec TPR Fb	RCV	Eval	Corr		3
SMOM (2017) (Zhu et al., 2017)	M	27 U	C45 Boost MLP NB	GM MAUC TPR Prec Fb PCR	RCV	Eval	Pair		3
AMDO (2018) (Yang et al., 2018)	M	17 KU	C45	Prec mP MAUC AUCm MS	RCV*	Fixed	Corr		5
k-means SMOTE (2018) (Douzas et al., 2018)	B	80 KU	LR kNN GB	GM F1 PRAUC	RCV*	Eval	Corr	yes	6
OBU (2018) (Vuttipittayamongkol et al., 2018)	B	36 KU	RF	TPR BA	HO*	Eval	Pair	yes	3
Buda et al. (2018) (Buda et al., 2018)	M	3	CNN	AUC Acc	FS*	Eval	–		2
CTGAN (2019) (Xu et al., 2019)	B+M	8 U	CART SVM MLP Boost LR	F1 mF1 Acc	FS*	NS	–	yes	3
G-SMOTE (2019) (Douzas and Bacao, 2019)	B	63 KU	LR kNN CART GB	AUC F1 GM	RCV*	Eval	Corr	yes	6
GAMO (2019) (Mullick et al., 2019)	M	7 KU	CNN	AvAcc GM	FS*	Eval	–	yes	3
RBO (2019) (Koziarski et al., 2019)	B	34 K	CART kNN SVM NB	Prec TPR Fb AUC GM	5x2*	Nest	Corr	yes	7
SOUP (2019) (Janicka et al., 2019)	M	15 U	C45 Rule kNN	GM PCR mF1	CV	Eval	Corr	yes	3
AB-SMOTE (2020) (Majzoub and Elgedawy, 2020)	B	8 KU	C45	TPR F1	CV	Eval	–		1
DLexiBoost (2020) (Datta et al., 2020)	B+M	33 K	C45 kNN	GM AUC MAUC	NS	Eval	Omni	yes	3
MBS (2020) (Liu and Hsieh, 2020)	B	13 U	LR Boost	AUC	RHO*	Eval	Pair	yes	4
MC-CCR (2020) (Koziarski et al., 2020)	M	20 K	C45 MLP kNN NB	AvAcc CBA GM CEN	RCV	Nest	Corr	yes	4
MC-RBO (2020) (Krawczyk et al., 2020)	M	20 K	C45 MLP kNN NB	AvAcc CBA GM CEN	RCV	Nest	Corr	yes	4
NBU (2020) (Vuttipittayamongkol and Elyan, 2020)	B	26 KU	SVM RF	TPR GM Prec F1	HO*	Fixed	Pair	yes	4
SPE (2020) (Liu et al., 2020)	B	5	kNN CART SVM MLP Boost RF GB LR C45	PRAUC F1 GM MCC TPR Prec	HO*	Fixed	–	yes	5
EMDO (2021) (Yao and Lin, 2021)	M	11 KU	C45	Prec mP AUC MAUC	RCV	NS	–		2
HDE (2021) (Chen et al., 2021a)	B	42 K	C45	AUC GM	RCV*	Eval	Corr	yes	5
RSMOTE (2021) (Chen et al., 2021b)	B+M	35 U	Boost LR CART kNN GB Oth	Prec TPR AUC F1 GM mF1 mP mR	RHO	Eval	Corr	yes	5
CBHSID (2022) (Palli et al., 2022)	B+M	16 KU	SVM	GM TPR F1	RHO	Fixed	Corr		3
FW-SMOTE (2022) (Maldonado et al., 2022)	B	42 KU	kNN SVM LR	AUC GM	CV*	Eval	Corr	yes	4
GDO (2022) (Xie et al., 2022)	B	52 KU	SVM	AUC GM	CV	NS	Pair		1

Study	Set.	Data	Learners	Measures	Prot.	Settings	Test	Code	Std.
KNNOR (2022) (Islam et al., 2022)	B	19 KU	SVM kNN CART MLP	AUC F1 GM	RCV	Eval	–	yes	4
SHSampler (2022) (Zheng et al., 2022)	M	20 K	CART RF NB	MAUC GM	CV	Fixed	Bayes		3
SOS (2022) (Kim et al., 2022)	B+M	6 U	CART LR Boost MLP	wF1 Acc	HO*	Eval	–	yes	3
SVDD-WSMOTE (2022) (Tao et al., 2022)	B	31 U	MLP LDA SVM	GM Fb AUC	RCV	Nest	Corr		4
GLOS (2023) (Han et al., 2023)	M	21 KU	C45	GM Prec PCR CBA	RCV	Eval	Pair		2
NROMM (2023) (Liu et al., 2023)	B+M	30 K	C45 MLP kNN NB SVM	AvAcc GM CBA CEN AUC Pmin mP AUCm MAUC	5x2*	Eval	Corr		4
OREM (2023) (Zhu et al., 2023)	B+M	49 U	C45 MLP SVM CART	F1 GM AUC mF1 MAUC TPR Prec BA	RCV	Nest	Pair		4
GMMSampling (2024) (Naglik and Lango, 2024)	M	11 U	CART NB kNN RF GB	BA mF1 GM MCC PRAUC	RCV*	Nest	Omni	yes	5
MC-MBRC (2024) (Ma et al., 2024)	M	20 KU	RF CART MLP kNN	AvAcc Fb CBA CEN	RCV	Eval	Corr	yes	3
MKC-SMOTE (2024) (Wang and Awang, 2024)	M	21 U	CART SVM	MAUC GM	CV	Eval	Pair		2
CISO (2025) (Yadav and Vijay, 2025)	B	50 KU	LR SVM RF GB	PRAUC AUC GM	CV*	Nest	Corr		5
GDHS (2025) (Yan et al., 2025b)	B+M	20 K	CART	GM MAUC	RCV	Eval	Bayes	yes	5
GOIO (2025) (Ren et al., 2025)	B	18 U	NB kNN CART SVM RF	F1 MCC Acc	RCV*	Fixed	Corr	yes	6
MC-NCLWO (2025) (Tao et al., 2025)	M	23 K	kNN CART NB SVM	Fb AvAcc CBA CEN	RCV	Eval	Corr	yes	4
MLOS (2025) (Yan et al., 2025a)	B	16 KU	SVM	AUC GM TPR	RCV	Eval	Bayes	yes	5
DGOT (2026) (Ren et al., 2026)	B+M	22 U	GB kNN CART LR RF	Acc F1 mF1 MCC	RCV*	Fixed	Corr	yes	6
Loyalty-SMOTE (2026) (Hu et al., 2026)	B+M	59 KU	SVM	F1 AUC TPR GM	HO	Fixed	–		2
QC-SMOTE (2026) (Upman and Gowda, 2026)	B	30 K	GB	AUC mF1	RCV*	Eval	Pair		2

Set.: B binary, M multi-class, B+M both. Data: number of real datasets; K KEEL, U UCI. Learners: C45 C4.5 or C5.0, CART, Rule rule learner, kNN, SVM, LR logistic regression, RF random forest, GB gradient boosting, Boost AdaBoost-type, NB naive Bayes, MLP, LDA, CNN, Oth other. Measures: Acc accuracy or error, TPR, TNR, Prec precision, FPR, GM G-mean, F1, Fb F-measure, Kap kappa, BA balanced accuracy, MCC, GS G-score, AUC ROC-AUC, PRAUC, MAUC, AUCm, PCR per-class recall, mF1 macro-F1, AvAcc, wF1 weighted F1, mP macro-precision, Pmin, CBA, CEN, mR macro-recall, MS minimum sensitivity, Cost. Prot.: RCV repeated k-fold CV, CV single k-fold CV, 5x2 CV, RHO repeated hold-out, HO single hold-out, FS fixed benchmark split, NS not stated; * train-only resampling stated. Settings: Eval chosen on evaluation data, Fixed fixed or default, Nest nested or held-out tuning, NS not stated. Test: Corr corrected multiple comparison, Pair pairwise without correction, Omni omnibus only, Bayes Bayesian, – none. Std.: points met of the standard in Table 29.

References

Abdi and Hashemi. To combat multi-class imbalanced problems by means of over-sampling techniques. *IEEE TKDE*, 2016. TODO: verify authors, pages and DOI.

Agrawal, Viktor, and Paquet. Scut: Multi-class imbalanced data classification using smote and cluster-based undersampling. 2015. TODO: verify authors, pages and DOI.

J. Alcalá-Fdez, L. Sánchez, S. García, et al. Keel: a software tool to assess evolutionary algorithms

for data mining problems. *Soft Computing*, 2009. doi: 10.1007/s00500-008-0323-y. TODO: verify volume and pages.

J. Alcalá-Fdez, A. Fernández, J. Luengo, et al. Keel data-mining software tool: Data set repository, integration of algorithms and experimental analysis framework. *J. Multiple-Valued Logic and Soft Computing*, 2011. TODO: verify volume and pages.

Barua, Islam, Yao, and Murase. Mwmote: Majority weighted minority oversampling technique for imbalanced data set learning. *IEEE TKDE*, 2014. TODO: verify authors, pages and DOI.

Batista, Prati, and Monard. A study of the behavior of several methods for balancing machine learning training data. In *SIGKDD Explorations*, 2004. TODO: verify authors, pages and DOI.

A. Benavoli, G. Corani, J. Demšar, and M. Zaffalon. Time for a change: a tutorial for comparing multiple classifiers through bayesian analysis. *JMLR*, 2017. TODO: verify volume and pages.

Rok Blagus and Lara Lusa. SMOTE for high-dimensional class-imbalanced data. *BMC Bioinformatics*, 14:106, 2013. doi: 10.1186/1471-2105-14-106.

Sabri Boughorbel, Fethi Jarray, and Mohammed El-Anbari. Optimal classifier for imbalanced data using Matthews correlation coefficient metric. *PLoS ONE*, 12(6):e0177678, 2017. doi: 10.1371/journal.pone.0177678.

Andrew P. Bradley. The use of the area under the ROC curve in the evaluation of machine learning algorithms. *Pattern Recognition*, 30(7):1145–1159, 1997. doi: 10.1016/S0031-3203(96)00142-2.

Dariusz Brzezinski and Jerzy Stefanowski. Prequential AUC: Properties of the area under the ROC curve for data streams with concept drift. *Knowledge and Information Systems*, 52(2):531–562, 2017.

Buda, Maki, and Mazurowski. A systematic study of the class imbalance problem in convolutional neural networks. *Neural Networks*, 2018. TODO: verify authors, pages and DOI.

Bunkhumpornpat, Sinapiromsaran, and Lursinsap. Safe-level-smote: Safe-level-synthetic minority over-sampling technique for handling the class imbalanced problem. In *PAKDD*, 2009. TODO: verify authors, pages and DOI.

Cao, Li, Woon, and Ng. Integrated oversampling for imbalanced time series classification. *IEEE TKDE*, 2013. TODO: verify authors, pages and DOI.

G. C. Cawley and N. L. C. Talbot. On over-fitting in model selection and subsequent selection bias in performance evaluation. *JMLR*, 2010. doi: 10.5555/1756006.1859921. TODO: verify volume and pages.

Chawla, Lazarevic, Hall, and Bowyer. Smoteboost: Improving prediction of the minority class in boosting. In *PKDD*, 2003. TODO: verify authors, pages and DOI.

Nitesh V. Chawla, Kevin W. Bowyer, Lawrence O. Hall, and W. Philip Kegelmeyer. SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16:321–357, 2002. doi: 10.1613/jair.953.

Nitesh V. Chawla, David A. Cieslak, Lawrence O. Hall, and Ajay Joshi. Automatically countering imbalance and its empirical relationship to cost. *Data Mining and Knowledge Discovery*, 17(2):225–252, 2008.

Chen, He, and Garcia. Ramoboost: Ranked minority oversampling in boosting. *IEEE TNN*, 2010. TODO: verify authors, pages and DOI.

Chen, Duan, Kang, and Qiu. A hybrid data-level ensemble to enable learning from highly imbalanced dataset. *Information Sciences*, 2021a. TODO: verify authors, pages and DOI.

Chen, Xia, Chen, Wang, and Wang. Rsmote: A self-adaptive robust smote for imbalanced problems with label noise. *Information Sciences*, 2021b. TODO: verify authors, pages and DOI.

Cieslak and Chawla. Start globally, optimize locally, predict globally: Improving performance on imbalanced data. In *ICDM*, 2008. TODO: verify authors, pages and DOI.

Cieslak, Chawla, and Striegel. Combating imbalance in network intrusion datasets. 2006. TODO: verify authors, pages and DOI.

Das, Krishnan, and Cook. Racog and wracog: Two probabilistic oversampling techniques. *IEEE TKDE*, 2015. TODO: verify authors, pages and DOI.

Datta, Nag, and Das. Boosting with lexicographic programming: Addressing class imbalance without cost tuning. *IEEE TKDE*, 2020. TODO: verify authors, pages and DOI.

J. Demšar. Statistical comparisons of classifiers over multiple data sets. *JMLR*, 2006. TODO: verify volume and pages.

J. Derrac, S. García, D. Molina, and F. Herrera. A practical tutorial on the use of nonparametric statistical tests as a methodology for comparing evolutionary and swarm intelligence algorithms. *Swarm and Evolutionary Computation*, 2011. doi: 10.1016/j.swevo.2011.02.002. TODO: verify volume and pages.

Diez-Pastor, Rodriguez, Garcia-Osorio, and Kuncheva. Random balance: Ensembles of variable priors classifiers for imbalanced data. *Knowledge-Based Systems*, 2015. TODO: verify authors, pages and DOI.

Douzas and Bacao. Geometric smote a geometrically enhanced drop-in replacement for smote. *Information Sciences*, 2019. TODO: verify authors, pages and DOI.

Douzas, Bacao, and Last. Improving imbalanced learning through a heuristic oversampling method based on k-means and smote. *Information Sciences*, 2018. TODO: verify authors, pages and DOI.

Fernandez, Lopez, Galar, del Jesus, and Herrera. Analysing the classification of imbalanced datasets with multiple classes: Binarization techniques and ad-hoc approaches. *Knowledge-Based Systems*, 2013. TODO: verify authors, pages and DOI.

Alberto Fernández, Salvador García, Francisco Herrera, and Nitesh V. Chawla. SMOTE for learning from imbalanced data: Progress and challenges, marking the 15-year anniversary. *Journal of Artificial Intelligence Research*, 61:863–905, 2018. doi: 10.1613/jair.1.11192.

Fernandez-Navarro, Hervas-Martinez, and Gutierrez. A dynamic over-sampling procedure based on sensitivity for multi-class problems. *Pattern Recognition*, 2011. TODO: verify authors, pages and DOI.

Freund and Schapire. A decision-theoretic generalization of on-line learning and an application to boosting. *JCSS*, 1997. TODO: verify authors, pages and DOI.

S. García and F. Herrera. An extension on "statistical comparisons of classifiers over multiple data sets" for all pairwise comparisons. *JMLR*, 2008. TODO: verify volume and pages.

S. García, A. Fernández, J. Luengo, and F. Herrera. Advanced nonparametric tests for multiple comparisons in the design of experiments in computational intelligence and data mining: Experimental analysis of power. *Information Sciences*, 2010. doi: 10.1016/j.ins.2009.12.010. TODO: verify volume and pages.

Han, Wang, and Mao. Borderline-smote: A new over-sampling method in imbalanced data sets learning. In *ICIC*, 2005. TODO: verify authors, pages and DOI.

Han, Guo, Li, and Wang. Global-local information based oversampling for multi-class imbalanced data. *IJMLC*, 2023. TODO: verify authors, pages and DOI.

D. J. Hand and R. J. Till. A simple generalisation of the area under the roc curve for multiple class classification problems. *Machine Learning*, 2001. doi: 10.1023/a:1010920819831. TODO: verify volume and pages.

Hassanat, Tarawneh, Altarawneh, and Almuhaimeed. Stop oversampling for class imbalance learning: A critical review. *Preprint (submitted to Elsevier, June 2022)*, 2022. TODO: verify authors, pages and DOI.

He, Bai, Garcia, and Li. Adasyn: Adaptive synthetic sampling approach for imbalanced learning. In *IJCNN*, 2008. TODO: verify authors, pages and DOI.

T. K. Ho and M. Basu. Complexity measures of supervised classification problems. *IEEE TPAMI*, 2002. doi: 10.1109/34.990132. TODO: verify volume and pages.

Hu et al. Loyalty-smote: Data synthesis algorithm for effective imbalanced data classification. *Neural Networks*, 2026. TODO: verify authors, pages and DOI.

Islam, Belhaouari, Rehman, and Bensmail. Knnor: An oversampling technique for imbalanced datasets. *Applied Soft Computing*, 2022. TODO: verify authors, pages and DOI.

Janicka, Lango, and Stefanowski. Using information on class interrelations to improve classification of multiclass imbalanced data: A new resampling algorithm. *Int. J. Appl. Math. Comput. Sci. (AMCS)*, 2019. TODO: verify authors, pages and DOI.

Kim, Lee, Shin, Park, et al. Sos: Score-based oversampling for tabular data. In *KDD*, 2022. TODO: verify authors, pages and DOI.

Koziarski, Krawczyk, and Wozniak. Radial-based oversampling for noisy imbalanced data classification. *Neurocomputing*, 2019. TODO: verify authors, pages and DOI.

Koziarski, Wozniak, and Krawczyk. Combined cleaning and resampling algorithm for multi-class imbalanced data with label noise. *Knowledge-Based Systems*, 2020. TODO: verify authors, pages and DOI.

Krawczyk, Koziarski, and Wozniak. Radial-based oversampling for multiclass imbalanced data classification. *IEEE TNNLS*, 2020. TODO: verify authors, pages and DOI.

Kubat and Matwin. Addressing the curse of imbalanced training sets: One-sided selection. In *ICML*, 1997. TODO: verify authors, pages and DOI.

Kubat, Holte, and Matwin. Machine learning for the detection of oil spills in satellite radar images. *Machine Learning*, 1998. TODO: verify authors, pages and DOI.

G. Lemaître, F. Nogueira, and C. K. Aridas. Imbalanced-learn: A python toolbox to tackle the curse of imbalanced datasets in machine learning. *JMLR*, 2017. TODO: verify volume and pages.

Liu and Hsieh. Model-based synthetic sampling for imbalanced data. *IEEE TKDE*, 2020. TODO: verify authors, pages and DOI.

Liu, Wu, and Zhou. Exploratory undersampling for class-imbalance learning. *IEEE TSMC-B*, 2009. TODO: verify authors, pages and DOI.

Liu, Liu, Yu, Zhong, and Hu. Noise-robust oversampling for imbalanced data classification. *Pattern Recognition*, 2023. TODO: verify authors, pages and DOI.

Liu et al. Self-paced ensemble for highly imbalanced massive data classification. In *ICDE*, 2020. TODO: verify authors, pages and DOI.

Victoria López, Alberto Fernández, Jose G. Moreno-Torres, and Francisco Herrera. Analysis of preprocessing vs. cost-sensitive learning for imbalanced classification. open problems on intrinsic data characteristics. *Expert Systems with Applications*, 39(7):6585–6608, 2012.

Victoria López, Alberto Fernández, and Francisco Herrera. On the importance of the validation technique for classification with imbalanced datasets: Addressing covariate shift when data is skewed. *Information Sciences*, 257:1–13, 2014.

V. López, A. Fernández, S. García, V. Palade, and F. Herrera. An insight into classification with imbalanced data: Empirical results and current trends on using data intrinsic characteristics. *Information Sciences*, 2013. doi: 10.1016/j.ins.2013.07.007. TODO: verify volume and pages.

Ma, Lu, and Jiang. A membership-based resampling and cleaning algorithm for multi-class imbalanced overlapping data. *Expert Systems with Applications*, 2024. TODO: verify authors, pages and DOI.

Al Majzoub and Elgedawy. Ab-smote: An affinitive borderline smote approach for imbalanced data binary classification. *IJMLC*, 2020. TODO: verify authors, pages and DOI.

Maldonado, Vairetti, Fernandez, and Herrera. Fw-smote: A feature-weighted oversampling approach for imbalanced classification. *Pattern Recognition*, 2022. TODO: verify authors, pages and DOI.

J. G. Moreno-Torres, J. A. Sáez, and F. Herrera. Study on the impact of partition-induced dataset shift on k -fold cross-validation. *IEEE TNNLS*, 2012. doi: 10.1109/tnnls.2012.2199516. TODO: verify volume and pages.

Mullick, Datta, and Das. Generative adversarial minority oversampling. In *ICCV*, 2019. TODO: verify authors, pages and DOI.

Naglik and Lango. Gmmsampling: a new model-based, data difficulty-driven resampling method for multi-class imbalanced data. *Machine Learning*, 2024. TODO: verify authors, pages and DOI.

Napierala, Stefanowski, and Wilk. Learning from imbalanced data in presence of noisy and borderline examples. In *RSCTC (LNAI 6086)*, 2010. TODO: verify authors, pages and DOI.

K. Napierała and J. Stefanowski. Types of minority class examples and their influence on learning classifiers from imbalanced data. *J. Intelligent Information Systems*, 2016. doi: 10.1007/s10844-015-0368-1. TODO: verify volume and pages.

Nekooeimehr. Oversampling methods for imbalanced dataset classification and their application to gynecological disorder diagnosis (ch. 3: Adaptive semi-unsupervised weighted oversampling (a-suwo) for imbalanced datasets). *PhD dissertation, University of South Florida (Ch. 3 reprints Expert Systems with Applications 2015, with Lai-Yuen)*, 2016. TODO: verify authors, pages and DOI.

Palli et al. A hybrid sampling approach for imbalanced binary and multi-class data using clustering analysis. *IEEE Access*, 2022. TODO: verify authors, pages and DOI.

Pradipta et al. Smote for handling imbalanced data problem: A review. In *IEEE conference (proceedings name not in PDF text)*, 2021. TODO: verify authors, pages and DOI.

Ren, Ding, Yang, and Cheung. Goio: Generative oversampling approach to class imbalance and overlap of tabular data. *IEEE TKDE*, 2025. TODO: verify authors, pages and DOI.

Ren, Ding, and Cheung. Diffusion gan-based oversampling for imbalanced tabular data. *IEEE TKDE*, 2026. TODO: verify authors, pages and DOI.

Saez, Luengo, Stefanowski, and Herrera. Smote-ipf: Addressing the noisy and borderline examples problem in imbalanced classification by a re-sampling method with filtering. *Information Sciences*, 2015. TODO: verify authors, pages and DOI.

T. Saito and M. Rehmsmeier. The precision-recall plot is more informative than the roc plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE*, 2015. doi: 10.1371/journal.pone.0118432. TODO: verify volume and pages.

Seiffert, Khoshgoftaar, Van Hulse, and Napolitano. Rusboost: A hybrid approach to alleviating class imbalance. *IEEE TSMC-A*, 2010. TODO: verify authors, pages and DOI.

Sun, Song, Zhu, Sun, et al. A novel ensemble method for classifying imbalanced data. *Pattern Recognition*, 2015. TODO: verify authors, pages and DOI.

Tao et al. Svdd-based weighted oversampling technique for imbalanced and overlapped dataset learning. *Information Sciences*, 2022. TODO: verify authors, pages and DOI.

Tao et al. Mc-nclwo: Newton's cooling law-based oversampling algorithm for multi-class imbalanced datasets with feature noise. *IEEE Access*, 2025. TODO: verify authors, pages and DOI.

Tomek. Two modifications of cnn. *IEEE TSMC*, 1976. TODO: verify authors, pages and DOI.

Luís Torgo, Rita P. Ribeiro, Bernhard Pfahringer, and Paula Branco. SMOTE for regression. In *Progress in Artificial Intelligence (EPIA 2013)*, volume 8154 of *Lecture Notes in Computer Science*, pages 378–389. Springer, 2013. doi: 10.1007/978-3-642-40669-0_33.

Upman and Gowda. Qc-smote: Quality-controlled smote for imbalanced classification. *arXiv 2606.24625 (Jun 2026)*, 2026. TODO: verify authors, pages and DOI.

Vuttipittayamongkol and Elyan. Neighbourhood-based undersampling approach for handling imbalanced and overlapped data. *Information Sciences*, 2020. TODO: verify authors, pages and DOI.

Vuttipittayamongkol, Elyan, Petrovski, and Jayne. Overlap-based undersampling for improving imbalanced data classification. In *IDEAL 2018 (LNCS 11341)*, 2018. TODO: verify authors, pages and DOI.

Wang and Awang. Mkc-smote: A novel synthetic oversampling method for multi-class imbalanced data classification. *IEEE Access*, 2024. TODO: verify authors, pages and DOI.

Wilson. Asymptotic properties of nearest neighbor rules using edited data. *IEEE TSMC*, 1972. TODO: verify authors, pages and DOI.

Wu, Xiong, Wu, and Chen. Local decomposition for rare class analysis. In *KDD*, 2007. TODO: verify authors, pages and DOI.

Xie, Qiu, Zhang, Peng, and Chen. Gaussian distribution based oversampling for imbalanced data classification. *IEEE TKDE*, 2022. TODO: verify authors, pages and DOI.

Xu, Skoularidou, Cuesta-Infante, and Veeramachaneni. Modeling tabular data using conditional gan. In *NeurIPS*, 2019. TODO: verify authors, pages and DOI.

Yadav and Vijay. Certified interpolation oversampling: Per-instance safety guarantees for imbalanced learning. *arXiv 2501.15790 (v2, Aug 2026)*, 2025. TODO: verify authors, pages and DOI.

Yan, Zheng, Han, Yu, and Zhou. Synthetic oversampling with mahalanobis distance and local information for highly imbalanced class-overlapped data. *ESWA*, 2025a. TODO: verify authors, pages and DOI.

Yan et al. Gdhs: An efficient hybrid sampling method for multi-class imbalanced data classification. *Neurocomputing*, 2025b. TODO: verify authors, pages and DOI.

Yang, Kuang, Zhang, and Zhang. Amdo: an over-sampling technique for multi-class imbalanced problems. *IEEE TKDE*, 2018. TODO: verify authors, pages and DOI.

Yao and Lin. Evolutionary mahalanobis distance-based oversampling for multi-class imbalanced data classification. *Sensors*, 2021. TODO: verify authors, pages and DOI.

Yen and Lee. Cluster-based under-sampling approaches for imbalanced data distributions. *ESWA*, 2009. TODO: verify authors, pages and DOI.

Zhang, Ma, Gan, Jiang, and Agam. Cgmos: Certainty guided minority oversampling. In *CIKM*, 2016. TODO: verify authors, pages and DOI.

Zheng, Yan, Zhang, and Zhang. Combating mutuality with difficulty factors in multi-class imbalanced data: A similarity-based hybrid sampling. 2022. TODO: verify authors, pages and DOI.

Zhou and Liu. On multi-class cost-sensitive learning. In *AAAI*, 2006. TODO: verify authors, pages and DOI.

Zhou and Liu. On multi-class cost-sensitive learning. *Computational Intelligence*, 2010. TODO: verify authors, pages and DOI.

Zhu, Lin, and Liu. Synthetic minority oversampling technique for multiclass imbalance problems. *Pattern Recognition*, 2017. TODO: verify authors, pages and DOI.

Zhu, Liu, and Zhu. Oversampling with reliably expanding minority class regions for imbalanced data learning. *IEEE TKDE*, 2023. TODO: verify authors, pages and DOI.